BOAMP Procurement Renewal Linking

Final Technical Report — Phase 1 and Phase 2

Detailed Reference for Data Audit, Preprocessing, Renewal Event Construction, and Survival Analysis

Internship Project — Gigalis

June 2026

Abstract

This report is the **detailed technical reference** for the internship work. It is intended to document the concrete implementation choices, data audit, preprocessing rules, intermediate diagnostics, renewal-linking logic, and survival-modeling outputs in a form that can be checked, reproduced, and handed off. BOAMP notices do not contain an explicit field identifying whether a call-for-tender (*appel d'offre*) renews a previous contract. The renewal relationship must therefore be inferred from observable signals: buyer identity, contract description, CPV classification, and timing relative to the expected contract end date.

Phase 1. Starting from 1,933 *APPEL_OFFRE* notices over 2015–2024, the pipeline defines 1,100 eligible source contracts and links 697 of them to a later renewal candidate, giving a final linking rate of 63.4%. The linking algorithm uses Sentence-Transformer embeddings with a four-component composite score. The official Phase-2 modeling input, `boamp_phase2_survival.csv`, contains one row per eligible source contract with a binary `event` indicator and an `observed_duration_months` survival time.

Phase 2. Survival analysis on the 1,100-contract dataset (697 events, 403 right-censored) yields a global Kaplan–Meier median survival of 48 months. The Cox proportional hazards model achieves an in-sample concordance of 0.6524 (C-index); the best-fitting parametric model is log-normal with AIC = 7 119.8 and concordance 0.6723. Statistically significant predictors are `declared_duration_months` (HR = 0.991, $p < 0.001$) and `start_year` (HR = 1.094, $p < 0.001$). A sensitivity analysis restricting to high-confidence links (composite $\geq$ 0.50, 494 events) yields a KM median of 52.7 months and Cox concordance of 0.6436, supporting the stability of the main qualitative conclusions under a stricter event definition. Finally, the survival model is translated into operational 12- and 24-month renewal-risk indicators at contract, buyer, and segment level — probabilities of an identifiable BOAMP renewal under the proxy-linking rule, intended as prioritisation indicators rather than legally verified renewal probabilities.

Executive Summary

This document is written for detailed verification. A reader who only wants a fast overview of the internship scope, workstreams, and headline results should start with the separate **internship report**. This technical report is the place for the exact mechanics.

The objective of Phase 1 is not to prove with legal certainty that two BOAMP notices are renewals of one another. The objective is to build a transparent, auditable, and reproducible proxy event variable for survival modeling. The final algorithm uses four signals: semantic similarity of the contract description, CPV compatibility, temporal proximity to the expected renewal date, and buyer identity.

The final method links 697 out of 1,100 eligible source contracts, or 63.4%. The gain comes mainly from two methodological changes: using Sentence-Transformer embeddings instead of lexical Jaccard/TF-IDF matching, and treating CPV as a scored compatibility signal rather than as a hard grouping constraint. The largest remaining weakness is buyer identification: only 12.8% of unique buyer entities are identified by SIRET, so most grouping relies on normalised buyer names. This can fragment the same public entity into several buyer keys and create false negatives.

The `event` variable should be interpreted carefully. `event = 1` means that the algorithm found a plausible renewal candidate. `event = 0` means that no candidate passed the linking rule in the observed data. It should not be read as proof that the contract was never renewed.

Building on the survival model, the report further translates the fitted log-normal AFT model into operational 12- and 24-month renewal-risk indicators at contract, buyer, and segment level (Section 13.10), giving Gigalis a prioritisation tool for renewal campaigns under the proxy-linking rule.

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1 Context and Objective

1.1 Business Context

Gigalis manages a portfolio of public digital infrastructure contracts in the Pays-de-la-Loire region. A critical operational question is: *when is a procurement contract likely to be renewed?* Knowing this in advance allows account managers to anticipate renewal campaigns, prepare competitive offers, and reduce the risk of losing contracts by default.

1.2 Why BOAMP?

The **Bulletin Officiel des Annonces de Marchés Publics** (BOAMP) is the official French public procurement gazette. All public contracts above the relevant thresholds must be published there, making it the most complete source of historical procurement data in France. It covers the full lifecycle: pre-information, call for tender (*appel d'offre*), award (*attribution*), and modification notices.

1.3 Core Problem

BOAMP does **not** provide an explicit field linking an original call for tender to its renewal. Each notice is a standalone record. The renewal relationship must therefore be **inferred** from a combination of signals: the same buyer re-issuing a contract for a similar service in the expected time-window after the original contract was due to expire.

1.4 Scope of Phase 1

Phase 1 produces a single BOAMP-only output table: one row per eligible *APPEL_OFFRE* with a binary column `event`. In this report, `event = 1` means that a plausible renewal was found by the linking algorithm, while `event = 0` means that no renewal link was observed under the chosen rule. The official Phase-2 handoff is `boamp_phase2_survival.csv`, where the event definition and censoring treatment are made explicit for survival modeling.

2 Raw Dataset

2.1 Source and Extraction

The dataset was extracted from the BOAMP API and covers notices published from **2015-03-02 to 2024-12-29**. The raw file after initial cleaning is `data/processed/boamp_full_clean.csv`.

The renewal linking algorithm operates exclusively on the 1,933 *APPEL_OFFRE* notices. ATTRIBUTION notices are used only as a secondary source to extract more precise contract start dates (via the `annonce_lie` back-reference field).

Table 1: Dataset overview

Notice type	Count	Share
APPEL_OFFRE (calls for tender)	**1,933**	**60.8%**
ATTRIBUTION (award notices)	1,081	34.0%
RECTIFICATIF (corrections)	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	**3,181**	100%

2.2 Column Dictionary — Raw Dataset (44 columns)

Table 2 describes all 44 columns in `boamp_full_clean.csv`. Columns are grouped by origin: raw BOAMP fields and fields added by the cleaning pipeline.

Table 2: Full column dictionary — `boamp_full_clean.csv`

Column	**Type**	**Coverage**	**Description**
— *Identifiers and dates* —			
`idweb`	string	100%	Unique BOAMP notice identifier (e.g. `22-111359`). Primary key.
`dateparution`	date	100%	Official publication date of the notice. Used as contract start proxy when no award date exists.
`datefindiffusion`	date	100%	End-of-diffusion date (administrative). Not used in modelling.
`datelimitereponse`	datetime	42.8%	Deadline for tender submission. Only partially filled.
`date_attribution`	date	0%*	Award date parsed from the notice body (0% on AO records; populated only on ATTRIBUTION notices).
`date_publication_anterieure`	date	0.4%	Reference to a prior publication. Sparse; not used.
— *Buyer information* —			
`nomacheteur`	string	100%	Raw buyer name as published. Inconsistent casing and accents.
`code_departement`	string	100%	French department code (e.g. `44` for Loire-Atlantique).
`buyer_siret`	string	9.1%	SIRET number (14-digit unique company ID). Low coverage.
`buyer_cp`	string	96.5%	Postal code of the buyer.
`buyer_ville`	string	99.6%	City name of the buyer.
`buyer_key`†	string	100%	**Constructed key**: `SIRET:XXXXXXXX` when SIRET is available, else `NAME:normalised_name`. Used as the grouping key for renewal matching.
— *Contract classification* —			
`famille`	string	100%	BOAMP family code (`JOUE` = EU Journal, or other).
`nature`	string	100%	Notice type (see Table 1 above).
`nature_libelle`	string	100%	Human-readable notice type label.

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Table 2 (continued)

Column	Type	Coverage	Description
`type_procedure`	string	98.5%	Procurement procedure: OUVERT (62.4%), PROCEDURE_ADAPTE (24.3%), NEGOCIE (8.9%), RESTREINT (1.9%), other.
`type_marche`	string	91.6%	Contract type: SERVICES (53.6%), FOURNITURES (32.0%), TRAVAUX (5.4%), mixed.
`type_avis`	string	95.1%	Notice subtype codes (e.g. `5|1|`). 100% on AO notices.
`etat`	string	100%	Publication state (`INITIAL` 97.6%, `ANNULE`, `RECTIFIE`).
`contractfolderid`	string	9.8%	Internal contract folder ID. Very sparse; not used.
— *Contract description* —			
`objet`	string	100%	Free-text contract description. Mean length: 105 characters (range: 6–201). The primary text signal for semantic matching.
`descripteur_libelle`	string	100%	Thematic descriptor in natural language (e.g. "Informatique (prestations de services)"). 549 unique values.
`titulaire`	string	0%	Winning contractor (populated only on ATTRIBUTION, not AO).
`annonce_lie`	string	2.9%	Back-reference to an earlier notice. On ATTRIBUTION records: 82.9% filled. Used as indirect external calibration signal (back-reference on ATTRIBUTION notices).
— *CPV codes* —			
`cpv_principal`	float	98.1%	Raw CPV code (8-digit integer) as published.
`cpv_all`	string	100%	Pipe-delimited list of all CPV codes for the notice.
`cpv_clean`†	float	98.1%	Cleaned primary CPV code. Generic codes (ending in `000000`) flagged separately.
`cpv_div2`†	float	98.1%	First 2 digits of CPV (division-level, e.g. 72 = IT services). Range: 15–98.
`cpv_class4`†	float	98.1%	First 4 digits of CPV (class-level granularity).
`category_id`†	string	100%	Internal category code (CAT01–CAT10, plus CAT_UNKNOWN) derived from CPV.
`category_label`†	string	100%	Human-readable category (see Table 3).
— *Contract amount* —			
`amount_eur`	float	43.0%	Raw amount in euros as parsed from the notice.
`amount_raw`	float	43.0%	Amount before cleaning transformations.
`amount_clean`†	float	41.1%	Cleaned amount: zero-values, suspiciously small values, and ceiling-flagged amounts removed. Range: €2,000 – €9.6M; median €350,000.
`flag_amount_zero`†	int	100%	1 if `amount_eur` = 0 (reported but missing).
`flag_amount_tiny`†	int	100%	1 if amount < €1,000 (likely misreported).
`flag_amount_ceiling`†	int	100%	1 if amount ≥ €10,000,000 (likely a framework ceiling, not actual spend).
— *Contract duration* —			
`duration_months`	float	47.7%	Duration in months as parsed from the notice.
`duration_raw_boamp`	float	47.7%	Raw parsed value before cleaning.
`duration_source_field`	string	47.7%	Source field from which duration was extracted (`DUREE_MOIS`, `DUREE_JOURS`, etc.).
`flag_duration_suspect`†	int	100%	1 if duration > 120 months (10 years) or = 0.
`duration_clean`†	float	47.3%	Cleaned duration: suspect values set to missing. Range: 1–120 months; median 24 months (among non-imputed).
— *Other* —			
`url_avis`	string	100%	Direct URL to the notice on boamp.fr.
`donnees_format`	string	100%	Data format flag (`legacy` 87.3%, `json` 12.7%).

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Table 2 (continued)

Column	**Type**	**Coverage**	**Description**

† Column added or derived by the cleaning pipeline. Not present in the original BOAMP API response.
* `date_attribution` is 0% for AO notices; it is populated on ATTRIBUTION notices and used indirectly via `annonce_lie`.

2.3 Contract Category Distribution

The 10 internal categories (CAT01–CAT10, plus an Unknown bucket for missing CPV) are derived from the CPV classification and cover exclusively the ICT/digital services scope of the Gigalis portfolio.

Table 3: Category distribution among APPEL_OFFRE notices

Category ID	Label	Count	Share
CAT02	IT Services & Consulting	583	30.2%
CAT01	Software & Applications	373	19.3%
CAT04	Telecom & Networks	337	17.4%
CAT03	Cybersecurity	211	10.9%
—	Unknown (CPV missing)	176	9.1%
CAT07	Digital Workplace & Collab.	89	4.6%
CAT08	Data & AI	54	2.8%
CAT06	IT Hardware & Equipment	45	2.3%
CAT10	IT Maintenance & Support	31	1.6%
CAT05	Cloud & Infrastructure	25	1.3%
CAT09	GIS & Mapping	9	0.5%
	Total AO	**1,933**	100%

2.4 Buyer Landscape

The dominance of name-based buyer keys (87.2%) is the primary structural risk: variant spellings of the same buyer (e.g. “Nantes” vs. “Nantes Métropole” vs. “NM”) fragment the buyer group, creating structural misses that the algorithm cannot recover.

2.5 Data Source Comparison: BOAMP vs. DECP

Table 5 provides a field-level comparison of both sources within scope (PdL digital contracts, 2015–2024). This informed the primary/enrichment architecture adopted for the rest of the analysis.

Table 4: Buyer concentration among APPEL_OFFRE notices

Metric	Value
Total unique buyers (`buyer_key`)	475
Buyers identified by SIRET	61 (12.8%)
Buyers identified by normalised name only	414 (87.2%)
Buyers with ≥ 2 AO notices	225 (47.4%)
Buyers with ≥ 5 AO notices	78 (16.4%)
Top 5 buyers by number of AO notices	
Nantes Métropole	128
Des Pays de la Loire	109
De Nantes	61
Saint-Nazaire	50
Angers Loire Métropole	46

Table 5: Field availability and completion by source ($n_{\text{BOAMP}} = 3{,}181$; $n_{\text{DECP}} = 3{,}039$)

Field	BOAMP		DECP		Recommended
	Avail.?	Fill%	Avail.?	Fill%	source
Buyer SIRET	Partial	9.1	Yes	100.0	DECP
Buyer name	Yes	100.0	Yes	100.0	Both
Contract object (text)	Yes	100.0	Yes	99.9	BOAMP
CPV code	Yes	98.1	Yes	100.0	Both
Amount (EUR)	Partial	43.0	Yes	95.6	DECP
Contract duration	Partial	47.7	Yes	99.4	DECP
Notification date	No	—	Yes	100.0	DECP
Publication date	Yes	100.0	Yes	98.0	BOAMP
Award date	Partial	31.8	No	—	BOAMP
Winner identity	Partial	27.6	Yes	95.5	Both
Procedure type	Yes	92.9	Yes	85.4	Both
Offers received	No	—	Partial	26.4	DECP
Notice type (AAPC/award)	Yes	100.0	No	—	BOAMP
Linked notices	Partial	34.9	No	—	BOAMP

Architecture decision. BOAMP was retained as the primary corpus: it is the only source covering the full 2015–2024 period, it distinguishes contract notices from award notices (the event structure survival analysis requires), and provides the richest free-text objects for Phase-

2 NLP classification. Although DECP was explored as a possible enrichment source, the current official handoff remains BOAMP-only because BOAMP and DECP do not share a robust direct contract-level linking key. DECP therefore remains a later enrichment path rather than part of the present modeling dataset.

2.6 Duration Field: Scientific Reliability Analysis

Scientific Question

Is the declared duration field (`duration_clean`) reliable as a survival target?

The answer is **no**. The declared duration is preserved as a covariate and as a temporal search prior for the linking algorithm, but it is not the survival target. The true survival target — the actual time elapsed between a contract start and its renewal — is the linked gap $\Delta(i, j^*) = t_{j^*} - t_i$, stored in `observed_duration_months` in the output file.

Evidence

Of the 1,933 *APPEL_OFFRE* notices, 1,480 (76.6%) have a valid, non-suspect `duration_clean` value. Table 6 shows the distribution of these values.

Table 6: Distribution of declared duration values (non-missing AO, $n = 1{,}480$)

Duration (months)	Count	Share	Cumulative
48	462	31.2%	31.2%
12	316	21.4%	52.6%
24	151	10.2%	62.8%
36	99	6.7%	69.5%
Subtotal: exactly 12/24/36/48 months	*1,028*	*69.5%*	
60	70	4.7%	74.2%
4	67	4.5%	78.7%
6	52	3.5%	82.2%
72	20	1.4%	83.6%
(other values)	243	16.4%	100.0%
Total	**1,480**	100%	

Mean: 30.8 months; Median: 24 months; Std: 21.2 months.

The diagnostic signal is the extreme clustering at round multiples of 12. Real contract durations driven by operational timelines would produce a continuous distribution with noise. Instead, nearly 70% of all values fall on exactly four integers — all multiples of 12 months

— and the next clear spikes are the administrative 5-year and 6-year ceilings (60 and 72 months), which again reflect legal maxima rather than empirically measured durations.

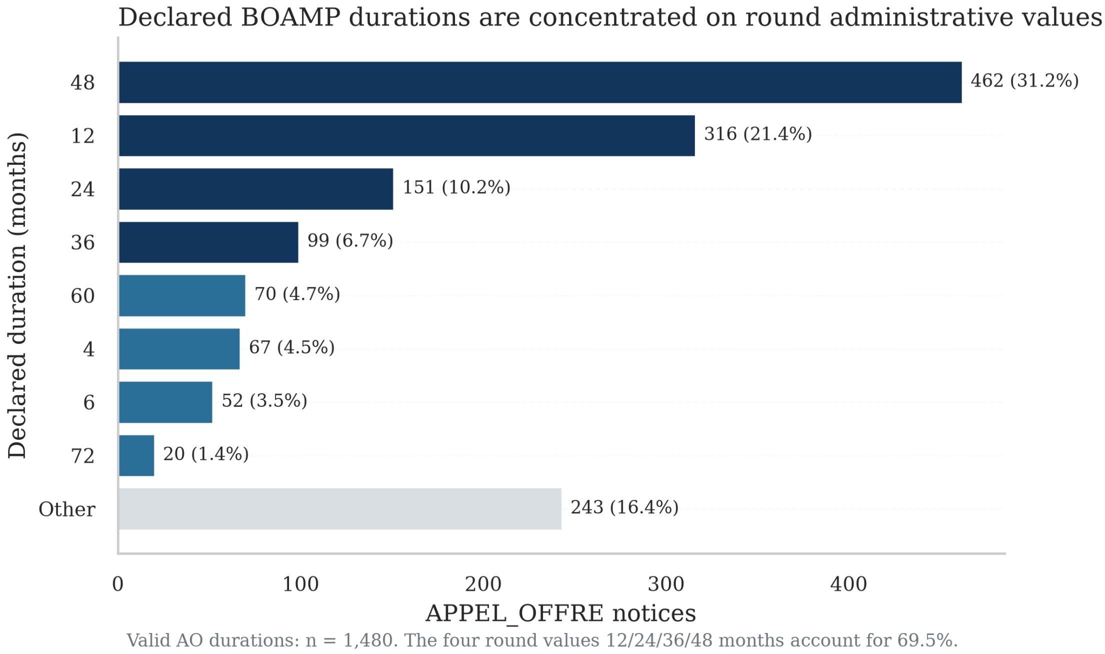


Figure 1: Distribution of declared duration values for the 1,480 *APPEL_OFFRE* notices with a valid `duration_clean`. Dark bars mark the four exact round values (12, 24, 36, 48 months) that together account for 69.5% of all valid durations. Light-blue bars are other specific values; grey is all remaining values combined. The concentration at round multiples of 12 reflects the administrative maximum declaration required by French procurement law, not empirically measured contract lifetimes.

Explanation

French public procurement law (Code de la commande publique) requires the buyer to declare the **administrative maximum duration** of the contract at the time of publication. This is defined as the base period plus all possible renewal tranches stacked end-to-end. A buyer writing a 12-month base contract with three possible 12-month renewal options declares `duration_months = 48`. The 48 months is a legal ceiling approved at signature, encoding the outer boundary of the contract, not a forecast of when the buyer intends to re-enter the market.

Using this field as the survival target would mean training the model on administrative ceilings rather than on observed behaviour. An AO that declares 48 months but renews after 12 months (at the end of the first base period) would be incorrectly labelled as a 48-month contract. The error is not random noise: it is systematically biased upward, because buyers default to the maximum configured option when completing the publication form.

Resolution: What the Field Is Used For

The declared duration d_i is retained for three specific purposes:

1. **Search prior in the linking algorithm.** The estimated end date $\hat{e}_i = t_i + d_i$ defines the centre of the temporal candidate window (Section 2.6 and Step 2 of the algorithm in Section 5). The declared duration tells the algorithm *where to look* for a renewal notice, not *what the answer is.*

2. **Covariate in survival models.** The administrative duration choice encodes the contractual structure the buyer selected (single-year vs. multi-year framework). This is predictive of renewal behaviour even when the raw value does not equal the actual time-to-renewal.

3. **Right-censoring indicator.** Contracts for which $\hat{e}_i = t_i + d_i > T_{\max}$ (December 2024) are excluded from the eligible denominator: their renewal window is not observable in the study period, regardless of algorithm quality.

The **actual survival target** is the observed gap:

$$T_i^{\text{obs}} = \Delta(i, j^*) = t_{j^*} - t_i \quad \text{(months)} \tag{1}$$

stored in `observed_duration_months` in `boamp_renewal_links.csv`. This quantity is available only for the 697 linked contracts (`event = 1`) and is the correct input duration for the Cox and Kaplan–Meier models in Phase 2. For right-censored contracts (`event = 0`), the survival model requires a censoring time, which is typically taken as the study end date minus the contract start date.

3 Dataset Construction and Missing-Value Control

This section explains the dataset in operational terms: what each row means, why each intermediate table exists, how the modeling population is constructed, and how incomplete fields are controlled. The objective is not only to document the pipeline, but also to make the dataset explainable during a technical discussion.

3.1 What Each Dataset Represents

The project does not start from a ready-made table of contract renewals. BOAMP publishes **notices**, not longitudinal contract histories. A BOAMP row is one publication event: a call for tenders, an award notice, a correction, or another administrative notice. The analysis therefore transforms a notice-level source into a contract-level survival dataset.

1. **Raw notice corpus.** The API extraction and flattening step produces `boamp_full_flat.csv` with 3,181 BOAMP notices and 30 raw/flattened columns.

2. **Cleaned analytical corpus.** The cleaning step produces `boamp_full_clean.csv` with 3,181 notices and 44 columns. This file keeps the raw variables and adds controlled fields such as `buyer_key`, `cpv_clean`, `amount_clean`, `duration_clean`, and taxonomy labels.

3. **Source contract population.** Only the 1,933 `APPEL_OFFRE` notices are used as possible source contracts, because they represent tenders that can later be re-tendered. Award, correction, and modification notices are informative context but are not survival units.

4. **Eligible modeling population.** A source contract enters the final linking denominator only if its expected renewal window is observable before the study end date, 2024-12-31. This gives 1,100 eligible contracts; the other 833 source notices are too recent to be evaluated fairly.

5. **Survival handoff.** The final file, `boamp_phase2_survival.csv`, has one row per eligible contract, with `event`, `observed_duration_months`, renewal diagnostics, and covariates used by Phase 2.

This distinction is important. The 3,181 rows describe the available BOAMP evidence; the 1,933 `APPEL_OFFRE` rows describe possible source contracts; the 1,100-row handoff is the population on which the official renewal-linking and survival results are measured.

3.2 How Missing Values Are Handled

The guiding rule is **flag first, preserve raw values, impute only when a downstream algorithm truly needs a value**. Missingness is treated as information about data quality, not as something to hide.

Table 7: Main missing-value and quality-control decisions

Field	**Observed quality**	**Handling rule**	**Reason**
Buyer identity	BOAMP SIRET is filled on 9.8% of AO notices; only 5.8% of AO rows become SIRET-anchored keys after validation.	Use `SIRET:...` when a valid 14-digit SIRET exists; otherwise use `NAME:...` from a normalised buyer name.	A weak name key is still necessary because renewal matching requires buyer grouping. The limitation is made explicit rather than discarded.
Contract amount	Only 21.4% of AO notices have a valid `amount_clean`; the full BOAMP corpus has 41.1% coverage.	Keep `amount_eur` and `amount_raw`; set `amount_clean` to missing when the value is zero, below €1,000, or at least €10M.	Zero and tiny amounts are likely encoding errors; very large values are often framework ceilings rather than actual spend. Amount is therefore not used as a core matching signal.
Contract duration	76.6% of AO notices have a clean declared duration; 5 AO durations are flagged as suspect.	Keep the raw duration; set suspect values outside 1–120 months to missing. For linking and survival-window construction only, missing duration is imputed to 48 months and `dur_was_imputed` records this choice.	The renewal window cannot be constructed without a duration. The 48-month value is a transparent default, not a claim that the true duration is known.
CPV code	96.8% of AO notices have a cleaned CPV code.	Derive the full CPV hierarchy (`cpv_full8`, `cpv_div2`, `cpv_group3`, `cpv_class4`, `cpv_category5`) and category labels when CPV exists. When CPV is missing in a candidate pair, it is excluded from the composite score and the remaining component weights are renormalized.	A missing CPV should not automatically erase a plausible renewal when text, buyer, and timing agree, but it must not inject an arbitrary neutral signal either.
Start date	AO notices do not directly carry a complete award-date field.	Use a linked ATTRIBUTION notice date when available; otherwise use `dateparution` as the start-date proxy.	Publication date is imperfect but observable for every notice; attribution back-references improve precision where they exist.
Renewal outcome	BOAMP has no explicit renewal label.	Build `event` by linking a later notice from the same buyer with compatible text, CPV, and timing. If no	`event=0` is not proof of no renewal; it means no renewal was observed under the algorithm and available

3.3 Practical Examples From Current Rows

Table 8 gives concrete examples from the current BOAMP handoff. These examples are useful because they show how the same rule is applied in different situations: a row can have missing amount, missing CPV, or an imputed duration and still be usable, provided the uncertainty is carried forward explicitly.

Table 8: Examples of raw-to-modeling transformations

Example	Raw evidence	Pipeline decision	Interpretation
Missing amount, still usable	`15-31464`: SDIS de la Mayenne, object about small fire and rescue equipment. Amount and CPV are missing; declared duration is 36 months.	`amount_clean = NaN`; `category_label = Unknown`; `duration_clean = 36`. The row remains eligible because buyer, text, and timing are still available.	Missing amount is a data-quality limitation, not a reason to delete the contract. Amount is not a core matching signal.
Imputed duration, explicitly flagged	`15-33241`: CHU de Nantes, acquisition and maintenance of an "appel malade" system. Raw duration is missing.	Duration is set to 48 months only for the renewal-window search, and `dur_was_imputed = True`. The row links to `20-19986` after 52.96 months.	The model can use the row, but the imputation flag tells us that the temporal prior is weaker than for an observed duration.
Censored contract, not bad data	`15-30947`: SMICTOM de la Vallee de l'Authion, software and equipment for waste collection. Amount is valid (€150,000) and declared duration is 12 months.	No accepted renewal is found. The hand-off stores `event = 0`, `renewal_duration_months = NaN`, `censoring_duration_months = 118.00`, and `observed_duration_months = 118.00`.	The row says: "we observed this contract for 118 months without an accepted renewal link." It is valid survival information.
SIRET fallback to name key	The same three examples have no valid BOAMP buyer SIRET.	The pipeline uses `NAME:de la mayenne`, `NAME:de nantes`, or `NAME:smictom de la vallee de l authion`.	This makes linking possible, but it also explains a risk: if the same buyer appears under another spelling, the buyer group may split.

Worked link example. For `15-31464`, the source notice is published on 2015-03-02 with a 36-month declared duration. The algorithm searches for later notices from the same buyer around the expected renewal period. It selects `17-176560` as the best candidate with `text_similarity = 0.6922`, `cpv_match_score` excluded (source CPV missing, `cpv_used_in_score = False`), `temporal_score = 0.7951`, and `composite_score = 0.7812`. The final survival row is therefore `event = 1` with `renewal_duration_months = 33.54`. This is a good example of why the method uses several signals: the CPV field is incomplete, but the buyer, text, and timing evidence are still strong enough to create a high-confidence link.

Worked censoring example. For `15-30947`, the contract starts on 2015-03-02 and has a clean 12-month declared duration and a valid amount. The algorithm does not find an accepted renewal candidate before 2024-12-31. The row is not removed; it becomes `event = 0` with a censoring duration of 118.00 months. In survival language, the only statement is: "the renewal time is longer than the observed follow-up, or the renewal was not detected." This is why the row remains in Kaplan–Meier and Cox modeling.

3.4 Why Right-Censoring Is Not Missing Data

The final dataset contains 697 linked renewals and 403 unlinked eligible contracts. The unlinked rows remain in the survival dataset because they still carry valid information: they survived without an observed renewal until the last date we could observe them. In survival-analysis notation, those rows have `event = 0` and a censoring duration. Removing them would bias the model toward contracts that renewed quickly and would make long-running or late-window contracts disappear from the risk set.

The interpretation is therefore:

- `event = 1`: the algorithm found a plausible later renewal notice; `renewal_duration_months` is observed.
- `event = 0`: no accepted renewal was observed before study end; `censoring_duration_months` is observed.
- `observed_duration_months`: always filled, using the renewal gap for events and the time-to-study-end for censored rows.

This is why the 1,100-row handoff is suitable for Kaplan–Meier and Cox models: every row contributes time-at-risk, even when the renewal event itself was not observed.

3.5 Practical Explanation of the Pipeline

In plain terms, the dataset is built as follows. First, the project collects all relevant digital-procurement BOAMP notices in Pays de la Loire from 2015 to 2024. Second, it cleans the unstable fields without overwriting the originals: buyer names become canonical keys, CPV codes become comparable groups, amounts and durations are flagged when implausible, and each contract receives a technology category. Third, it keeps only call-for-tender notices as contracts that can later be renewed. Fourth, it searches forward in time for later notices from the same buyer that look like the same service being re-tendered. Finally, it writes a survival table where each contract has a start date, a duration, an event indicator, and enough diagnostics to judge whether the inferred renewal link is strong or weak.

The most important limitation to remember is that the target variable is constructed, not directly observed. The strength of the project is that this construction is explicit: each assumption is visible in the output columns and can be audited, adjusted, or tested through sensitivity analysis.

4 Data Quality Issues Diagnosed

Nine data quality issues were identified through systematic EDA. Each directly affects the accuracy or recall of the renewal linking algorithm.

Table 9: Data quality issues and their impact on the linking algorithm

#	Issue	Observation	Impact on linking
1	SIRET coverage low	Only 5.8% of *APPEL_OFFRE* notices have a SIRET-anchored `buyer_key`. The SIRET *field* is present on 9.8% of AO (9.1% across all 3,181 notices), but ∼46% of those values fail validation and fall back to the name key; 12.8% of the 475 unique buyer keys are SIRET-anchored. The rest use normalised name as buyer key	Variant spellings split buyer groups ⇒ structural misses
2	Buyer name variants	Same buyer appears under multiple spellings after normalisation	False negative links where buyers are split
3	CPV missing or generic	3.2% of AO have no CPV; generic catch-all codes are capped at the division level	Missing CPV is excluded from the composite and the remaining weights renormalized
4	Contract amount sparse	Only 21.4% of AO report a valid, non-flagged amount	Amount cannot be used as a matching signal
5	Duration missing	23.4% of AO have no duration; imputed to 48 months	Wrong imputed duration shifts the temporal window
6	Duration suspect values	Some raw durations > 120 months or $= 0$	Removed; fall into imputation bucket
7	Contract start date imprecise	`date_attribution` is 0% on AO records; publication date used as proxy for 59.3%	Up to several weeks of noise in temporal scoring
8	Short or generic *objet*	24 notices have `objet` < 20 characters	Semantic embeddings less discriminative for short texts
9	`annonce_lie` interpretation	Present only on ATTRIBUTION notices (82.9%); refers to same-contract back-link, not a renewal	Used as calibration signal, not as a direct renewal label

5 Preprocessing Pipeline

5.1 Buyer Key Construction

$$\texttt{buyer\_key}_i = \begin{cases} \texttt{SIRET:}s_i & \text{if SIRET } s_i \text{ is available} \\ \texttt{NAME:}\phi(n_i) & \text{otherwise} \end{cases} \tag{2}$$

where $\phi(\cdot)$ is the name normalisation function: lowercase, strip accents, remove punctuation, collapse whitespace.

5.2 Contract Start Date Refinement

Award date from linked ATTRIBUTION notices is used when available, falling back to the publication date:

$$t_i = \begin{cases} \text{award\_date}(j) & \text{if } \exists\, j \in \text{ATTR} : \texttt{annonce\_lie}(j) = i \\ \texttt{dateparution}_i & \text{otherwise} \end{cases} \tag{3}$$

In practice, 40.7% of eligible AO obtain a more precise start date through this refinement.

5.3 Duration Imputation

$$d_i = \begin{cases} \texttt{duration\_clean}_i & \text{if not missing and not suspect} \\ d_0 = 48 \text{ months} & \text{otherwise (23.4\% of AO)} \end{cases} \tag{4}$$

5.4 Estimated End Date

$$\hat{e}_i = t_i + d_i \tag{5}$$

5.5 Text Normalisation

The *objet* field is normalised for TF-IDF (used in calibration) and passed raw to Sentence-Transformers:

1. Lowercase
2. Strip Unicode accents (NFD normalisation)
3. Keep only characters matching `[a-z0-9 ]`
4. Collapse multiple spaces

5.6 Sentence-Transformer Embeddings

Each normalised *objet* $\mathbf{o}_i$ is encoded into a 384-dimensional unit-norm vector:

$$\mathbf{v}_i = \mathrm{ST}\big(\mathbf{o}_i\big) \in \mathbb{R}^{384}, \qquad \|\mathbf{v}_i\|_2 = 1 \tag{6}$$

Model used: `paraphrase-multilingual-MiniLM-L12-v2` (multilingual, 384 dimensions, ∼120 MB).

5.7 Cleaning Statistics Summary

Tables 10–12 quantify the impact of the Week-2 cleaning rules applied to both sources.

Table 10: Amount cleaning flags and fill rates by source

Source	Zero (= 0)	Tiny (<€1 k)	Ceiling (≥€10 M)	`amount_clean` fill
BOAMP ($n = 3{,}181$)	1	9	53	41.1%
DECP ($n = 3{,}039$)	31	20	23	93.1%

Flagged values are set to `NaN` in `amount_clean` but the raw field is preserved. The low BOAMP fill rate reflects the structural gap documented in Table 5: amount reporting was mandatory only for eForms notices (2024+). No amount imputation is applied; median-by-segment imputation is deferred to Phase 3 modelling.

Table 11: Duration cleaning and fill rates by source

Source	Suspect flags (outside [1, 120] months)	`duration_clean` fill
BOAMP ($n = 3{,}181$)	12	47.3%
DECP ($n = 3{,}039$)	6	99.2%

Table 12: Buyer key normalization results

Source	Raw unique names	Canonical keys	SIRET-anchored
BOAMP	525	502	156 rows (4.9%)
DECP	278	294	3,039 rows (100.0%)

The cross-source buyer bridge is saved to `buyer_bridge.csv`. Name normalisation applies lowercase, accent stripping, punctuation removal, and legal-form suffix removal (e.g. *Commune de*, *SARL*).

6 Renewal Linking Algorithm

6.1 Notation

Let $\mathcal{A}$ denote the set of all *APPEL_OFFRE* notices. Each notice $i \in \mathcal{A}$ is represented by:

- t_i: contract start date, defined as the award date when available, otherwise the BOAMP publication date;
- d_i: contract duration in months, after cleaning and imputation;
- $\hat{e}_i = t_i + d_i$: estimated contract end date;
- CPV_i: main CPV classification code;
- $\mathbf{o}_i$: contract description text (*objet*);
- b_i: buyer identifier, using SIRET when available and normalised buyer name otherwise.

The goal is to construct, for each eligible source notice i, an algorithmic binary outcome Y_i indicating whether a plausible renewal candidate can be found among later BOAMP calls-for-tender.

6.2 Step 1 — Eligibility Filter

A notice is **eligible** to be a renewal source if its estimated renewal date falls within the study period $[T_{\min}, T_{\max}]$:

$$i \in \mathcal{A}_{\text{elig}} \iff T_{\min} \leq \hat{e}_i \leq T_{\max} \quad \big(T_{\min} = \text{2015-01-01},\ T_{\max} = \text{2024-12-31}\big) \tag{7}$$

Table 13: Eligibility breakdown

Population	Count
Total APPEL_OFFRE	1,933
Not eligible: estimated end date after 2024	833
Eligible ($\mathcal{A}_{\text{elig}}$)	**1,100**

Contracts with $\hat{e}_i > T_{\max}$ are excluded from the Phase 1 linking denominator because the renewal opportunity is not sufficiently observable in the 2015–2024 dataset.

6.3 Step 2 — Candidate Pool

For each eligible source i, the candidate pool is:

$$\mathcal{C}(i) = \big\{ j \in \mathcal{A} : b_j = b_i, \quad t_j > t_i, \quad |t_j - \hat{e}_i| \leq W \big\}, \quad W = 12 \text{ months} \tag{8}$$

The gap between source and candidate is:

$$\Delta(i, j) = t_j - t_i \quad \text{(in months)} \tag{9}$$

Key design choice: candidates are grouped by `buyer_key` only. The previous baseline grouped by `[buyer_key, cpv_div2]`, which rejected valid renewals where the buyer reclassified the service under a different CPV code.

6.4 Step 3 — Component Scores

Text similarity. Cosine similarity between unit-norm embeddings (dot product = cosine when both vectors are ℓ_2-normalised):

$$s_{\text{text}}(i, j) = \mathbf{v}_i \cdot \mathbf{v}_j \tag{10}$$

In theory, cosine similarity lies in $[-1, 1]$. In this application, the hard text-similarity filter below retains only candidate pairs with $s_{\text{text}}(i, j) \geq 0.20$.

CPV compatibility. A hierarchy-based score following the official CPV nomenclature, in which a code is read left-to-right as division (2 digits), group (3), class (4), category (5) and full code (8); a 9th digit after a dash is only a check digit. The score rewards the depth of the longest shared prefix. A CPV code is treated as generic if it ends in `000000` (e.g. `72000000` for IT services in general); such codes carry no sub-division signal and are capped at the division level. A missing CPV is *excluded* from the composite (returned as `NaN`) rather than scored, and the remaining weights are renormalized (Section 6.5).

$$s_{\text{cpv}}(i, j) = \begin{cases} \text{NaN} & \text{either CPV missing (excluded, renormalized)} \\ 0.20 & \text{either generic \& same division} \\ 0.00 & \text{either generic \& different division} \\ 1.00 & \text{CPV}[:8]_i = \text{CPV}[:8]_j \text{ (same full code)} \\ 0.80 & \text{CPV}[:5]_i = \text{CPV}[:5]_j \text{ (same category)} \\ 0.60 & \text{CPV}[:4]_i = \text{CPV}[:4]_j \text{ (same class)} \\ 0.40 & \text{CPV}[:3]_i = \text{CPV}[:3]_j \text{ (same group)} \\ 0.20 & \text{CPV}[:2]_i = \text{CPV}[:2]_j \text{ (same division)} \\ 0.00 & \text{otherwise (valid but different division)} \end{cases} \tag{11}$$

Temporal proximity. Linear decay from 1.0 at the exact expected renewal date to 0.0 at the window boundary $W = 12$ months:

$$s_{\text{temp}}(i,j) = \max\left(0,\ 1 - \frac{|\Delta(i,j) - d_i|}{W}\right) \tag{12}$$

Buyer match. Always 1.0 within the same-buyer grouping loop:

$$s_{\text{buyer}}(i,j) = 1.0 \tag{13}$$

Buyer identity is therefore mainly a *blocking rule*: candidates are first restricted to the same buyer group. The constant buyer score does not change the ranking among candidates for the same source contract, but it is kept explicit to document the full matching design and to allow future extension to cross-buyer matching.

6.5 Step 4 — Composite Score

When the CPV code is present on both notices, the four components are combined with fixed weights:

$$\boxed{S(i,j) = 0.40 \cdot s_{\text{text}} + 0.25 \cdot s_{\text{cpv}} + 0.20 \cdot s_{\text{temp}} + 0.15 \cdot s_{\text{buyer}}} \tag{14}$$

When the CPV code is missing on either notice, CPV is excluded and the remaining component weights are renormalized by their sum ($0.40 + 0.20 + 0.15 = 0.75$) so the score stays in $[0,1]$:

$$S(i,j) = \frac{0.40 \cdot s_{\text{text}} + 0.20 \cdot s_{\text{temp}} + 0.15 \cdot s_{\text{buyer}}}{0.75}. \tag{15}$$

Missing CPV is therefore never assigned an arbitrary neutral 0.5. A boolean `cpv_used_in_score` column records, per pair, whether CPV contributed.

6.6 Step 5 — Hard Filters

Two hard gates are applied before best-match selection:

$$s_{\text{text}}(i,j) \geq 0.20 \tag{16}$$

$$\Delta(i,j) \in [6,\ 72] \text{ months} \tag{17}$$

The minimum gap of 6 months prevents matching a contract to itself or to an immediate amendment. The maximum gap of 72 months (6 years) covers contracts with the longest expected durations.

Table 14: Weight rationale

Component	Weight	Rationale
s_{text}	0.40	Strongest discriminator. Semantic similarity captures service identity beyond lexical overlap. Calibration shows ST cosine ≥ 0.30 for 96% of verified same-contract pairs.
s_{cpv}	0.25	Structured and reliable when specific. Degrades gracefully through the CPV hierarchy (full code $\to$ category $\to$ class $\to$ group $\to$ division); generic codes capped at the division level (0.20). Missing CPV is excluded and the remaining weights renormalized. Not used as a hard gate because buyers sometimes reclassify.
s_{temp}	0.20	Rewards timing alignment without hard-excluding distant candidates (hard filters apply separately).
s_{buyer}	0.15	Constant within the loop; kept explicit so all weights sum to 1.0. Included for future extension to cross-buyer matching.

CPV is intentionally **not** used as a hard filter. Requiring CPV agreement as a gate would exclude legitimate renewals when the same buyer reclassifies a similar service under a different CPV code.

Let $\mathcal{F}(i)$ denote the filtered candidate set:

$$\mathcal{F}(i) = \big\{ j \in \mathcal{C}(i) : s_{\text{text}}(i,j) \geq 0.20,\ \Delta(i,j) \in [6,72] \big\}. \tag{18}$$

6.7 Step 6 — Best-Match Selection

For each source, keep the single highest-scoring candidate:

$$j^*(i) = \underset{j \in \mathcal{F}(i)}{\arg\max}\ S(i,j) \tag{19}$$

where $\mathcal{F}(i) \subseteq \mathcal{C}(i)$ is the subset passing both hard filters (16) and (17).

6.8 Step 7 — Outcome Variable

The Phase 1 binary outcome is an *algorithmic linking outcome*. It records whether the matching rule found a plausible renewal candidate for source contract i.

$$Y_i = \begin{cases} 1 & \text{if } \mathcal{F}(i) \neq \emptyset \quad \text{(renewal found)} \\ 0 & \text{if } \mathcal{F}(i) = \emptyset \quad \text{(no renewal link observed)} \end{cases} \tag{20}$$

The final linking rate is:

$$\text{Linking rate} = \frac{\sum_{i \in \mathcal{A}_{\text{elig}}} Y_i}{|\mathcal{A}_{\text{elig}}|} = \frac{697}{1{,}100} = 63.4\%. \tag{21}$$

The value $Y_i = 0$ should be interpreted as an unobserved link under the available data and chosen thresholds, not as definitive evidence that no renewal occurred.

7 Output Dataset — Modeling Input

7.1 File: `boamp_phase2_survival.csv`

This is the official BOAMP-only Phase-2 handoff. It is exported from the notebook linking output and contains one row per eligible *APPEL_OFFRE* (1,100 rows, 32 columns).

Table 15: Column dictionary — `boamp_phase2_survival.csv` (modeling input)

Column	Type	Description
— *Identifiers* —		
`contract_id`	string	Source contract BOAMP ID (e.g. `BOAMP:15-31464`).
`renewal_contract_id`	string	Best-matched renewal contract ID. Empty when `event=0`.
— *Buyer and classification* —		
`source`	string	Constant value `BOAMP`; makes the handoff schema explicit.

Table 15 (continued)

Column	Type	Description
`buyer_key`	string	Buyer identifier (`SIRET:...` or `NAME:...`). 308 unique buyers. In practice every row in the final eligible cohort carries a `NAME:`-based key: no SIRET was populated for any of the 1,100 eligible digital-sector AO contracts, so the `NAME:` fallback applied universally and name-based grouping governs the entire modeled cohort (and its full candidate pool). This is structural, not incidental: every SIRET-anchored AO is a 2024 eForms notice, and all 2024 contracts are censored upfront (their estimated renewal window falls after the 2024-12-31 study end), so SIRET coverage in the eligible cohort is necessarily zero. SIRET enrichment can therefore only help future (post-2024) data, not the historical survival window.
`cpv_div2`	float	2-digit CPV division of the source contract. 95.3% coverage.
`category_label`	string	Internal category (11 levels; see Table 3).
— *Temporal variables* —		
`start_date`	date	Source contract start date (refined from award date where available).
`declared_duration_months`	float	Contract duration as declared (or imputed). Mean: 31.5 months.
`dur_was_imputed`	bool	True if duration was set to the default 48-month imputation (25.1% of rows).
`estimated_end_date`	date	$\hat{e}_i = t_i + d_i$; defines the renewal search window.
`observed_duration_months`	float	Survival time in months, always filled. For `event=1`: observed gap to the linked BOAMP renewal candidate $\Delta(i, j^*)$. For `event=0`: right-censoring duration from contract start to end of study period.
`renewal_duration_months`	float	Observed renewal gap for linked contracts only. Filled on 63.4% of rows (the `event=1` subset).
`censoring_duration_months`	float	Observed time-to-study-end for censored contracts only. Filled on 36.6% of rows (the `event=0` subset).
— *Outcome variable* —		
`event`	int	**Target variable**. 1 = plausible renewal link found (697), 0 = no renewal link observed under the algorithm (403).

Table 15 (continued)

Column	Type	Description
— Matching scores (linked rows only) —		
`composite_score`	float	$S(i, j^*)$. Mean: 0.579, range: 0.245–0.998.
`text_similarity`	float	$s_{\text{text}}(i, j^*)$. Mean: 0.517, range: 0.204–1.000.
`cpv_match_score`	float	$s_{\text{cpv}}(i, j^*)$. Mean: 0.301, range: 0.000–1.000 (over the 646 links with a CPV; NaN for the 51 missing-CPV links).
`cpv_used_in_score`	bool	True when CPV contributed to S (646 links); False when CPV was missing and the weights were renormalized (51 links).
`temporal_score`	float	$s_{\text{temp}}(i, j^*)$. Mean: 0.706, range: 0.006–1.000.
— Metadata —		
`start_date_source`	string	How t_i was obtained: `publication_date` (59.3%) or `attribution_award_date` (40.7%).
`link_method`	string	Algorithm used: `sentence_transformers` (697 rows) or `none` (403 rows).
— Covariates (joined from `boamp_full_clean.csv`) —		
`amount_clean`	float	Contract amount (EUR) after quality filtering; NaN when any amount flag is True. Fill rate: 20.3% (BOAMP-only limitation).
`flag_amount_zero`	bool	True if raw amount = 0 (encoding artefact).
`flag_amount_tiny`	bool	True if raw amount $\in (0, 1\,000)$ EUR (implausible for an IT contract).
`flag_amount_ceiling`	bool	True if raw amount $\geq 10\,000\,000$ EUR (likely framework-agreement ceiling).
`type_procedure`	string	Procurement procedure type (e.g. *Procédure adaptée*, *Appel d'offres ouvert*). Fill rate: 99.7%.
`type_marche`	string	Contract type (e.g. *Marché*, *Accord-cadre*). Fill rate: 85.6%.
— Link confidence diagnostics (added in the Phase-2 handoff) —		
`best_composite_score`	float	$S(i, j^{(1)})$: composite score of the best candidate (the selected link for `event`=1).
`second_best_composite_score`	float	$S(i, j^{(2)})$: composite score of the runner-up. NaN for single-candidate sources.
`score_margin`	float	$\Delta = S(i, j^{(1)}) - S(i, j^{(2)})$. Best-minus-second-best gap; NaN when only one candidate.

Table 15 (continued)

Column	**Type**	**Description**
`n_candidates_for_source`	int	Number of filtered candidates for the source (135 sources have exactly one; 562 have ≥ 2).
`single_candidate_match`	bool	True when the source had a single candidate (135 rows); margin undefined, not auto-promoted to strict.
`high_confidence_strict`	bool	True iff `event`=1 $\wedge$ $S \geq 0.70 \wedge \Delta \geq 0.05$ (90 rows). Conservative sensitivity flag for Model C (Section 11.2).

7.2 Score Distributions for Linked Contracts

Table 16 reports the distribution of matching scores among the 697 linked contracts. These statistics are useful diagnostics, but they should not be interpreted as independent validation: the same scores are used by the algorithm to select the links. The low minimum values, especially for the composite, CPV, and temporal scores, indicate that some accepted links are low-confidence cases that should be checked in the Phase 2 validation sample.

Table 16: Descriptive statistics of matching scores (event=1 rows only, n=697; `cpv_match_score` over the 646 links with a non-missing CPV)

Score	Mean	Std	Min	Q25	Median	Q75	Max
`composite_score`	0.579	0.141	0.245	0.489	0.555	0.653	0.998
`text_similarity`	0.517	0.200	0.204	0.367	0.475	0.619	1.000
`cpv_match_score`	0.301	0.366	0.000	0.000	0.200	0.400	1.000
`temporal_score`	0.706	0.253	0.006	0.587	0.775	0.913	1.000

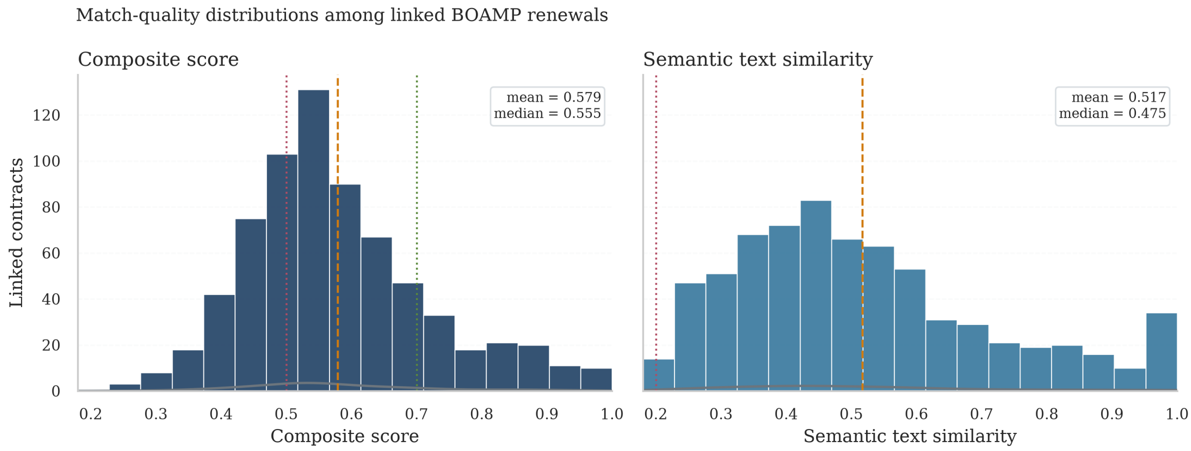


Figure 2: Score distributions for the 697 linked contracts (`event = 1`). **Left:** composite score $S(i, j^*)$ with confidence-tier shading: LOW (composite < 0.50, red, $n = 203$), MEDIUM (0.50–0.70, orange, $n = 372$), HIGH (≥ 0.70, green, $n = 122$). The dashed line marks the mean (0.579). **Right:** text similarity $s_{\text{text}}(i, j^*)$ with the hard-filter threshold at 0.20 (dotted red). The shaded band (0.20–0.30) flags 81 pairs (11.6%) whose semantic similarity is low, representing the highest false-positive risk.

7.3 Declared vs. Observed Duration

Table 17 compares the declared contract duration with the gap between the source notice and the selected renewal candidate. This is a consistency check for the temporal matching logic, not external proof that the selected pairs are true renewals, because declared duration is already used to define the candidate window and the temporal score.

Table 17: Duration statistics (months)

	Mean	Std	Min	Q25	Median	Q75	Max
`declared_duration_months` (all 1,100)	31.5	17.7	1	12	36	48	72
`observed_duration_months` (697 linked)	32.3	17.0	6	13	37	48	71

The close medians, 36 months for declared duration and 37 months for the linked candidate gap, are therefore reassuring but partly mechanical. A stronger validation should examine the error `observed_duration_months` − `declared_duration_months`, separately for imputed and non-imputed durations, and combine this with manual review of a stratified sample of linked pairs.

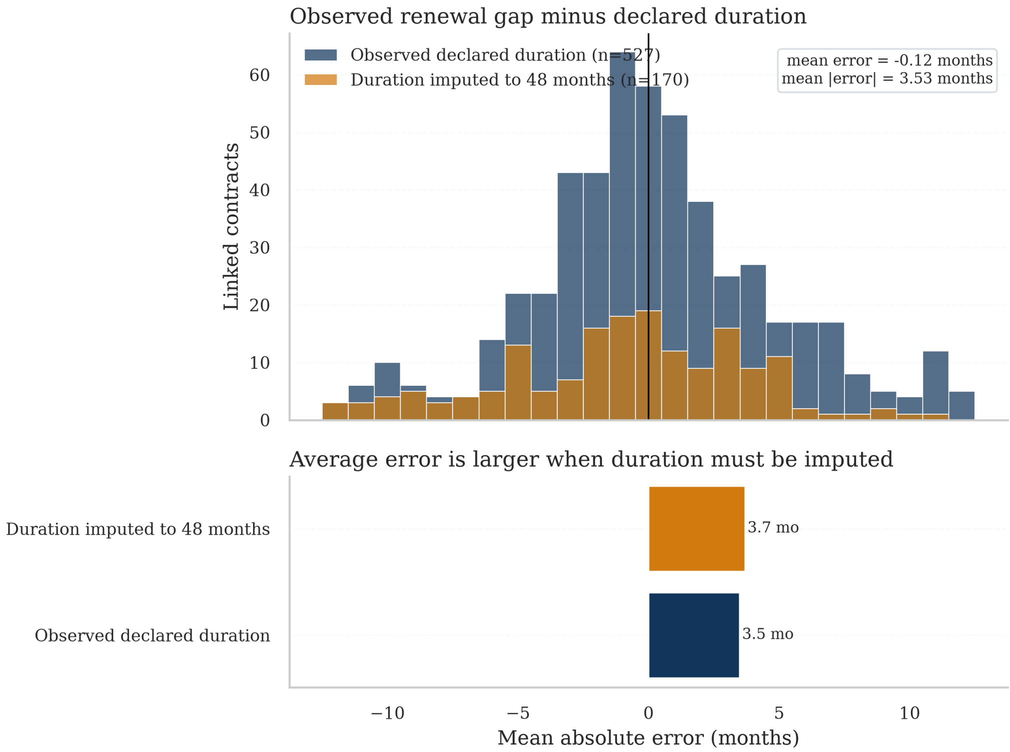


Figure 3: Distribution of the temporal error $\Delta(i, j^*) - d_i$ (observed gap minus declared duration) for the 697 linked contracts. Blue bars correspond to the 527 contracts with a valid non-imputed duration; orange bars to the 170 contracts whose duration was imputed to 48 months. The overall mean error is -0.04 months (effectively zero bias) with a standard deviation of 4.87 months. All pairs fall within ± 12 months by construction of the hard filter, but the near-zero mean is consistent with the absence of a systematic temporal shift in the matches, though it cannot rule out residual bias from unobserved confounders. Imputed-duration contracts show slightly larger spread (mean $|\text{error}| = 4.1$ vs. 3.7 months for non-imputed).

8 Results

8.1 Linking Rate Comparison

Table 18: Linking rate by method

Method	Linked	Denominator	Rate
Baseline (Jaccard + buyer × CPV groupby)	219	1,933	11.3%
TF-IDF cosine (intermediate)	279	1,100	25.4%
Sentence-Transformers (final)	**697**	**1,100**	**63.4%**

The two key changes responsible for the improvement are:

1. **Grouping by buyer only** (removing CPV as a hard gate): recovers contracts where

the buyer reclassified the service. The final method also reports performance on the 1,100 contracts whose expected renewal date is observable within the study period, instead of using all 1,933 AO notices as the denominator.

2. **Sentence-Transformers over TF-IDF**: semantic similarity captures meaning beyond lexical overlap, contributing approximately +38 percentage points in linking rate compared to TF-IDF alone.

8.2 Failure Analysis

Of the 403 unlinked eligible AO, three failure modes are identified:

Table 19: Failure mode breakdown (403 unlinked eligible AO)

Failure reason	Count	Share	Description
`NO_TEMPORAL_PARTNER`	339	84.1%	No other AO from the same buyer fell in the ±12-month window. Likely structural (buyer issued only one contract in the period).
`TEXT_MISMATCH`	32	7.9%	Temporal candidates existed but all had $s_{\text{text}} < 0.20$. Text was too dissimilar even at the semantic level.
`CPV_MISMATCH`	32	7.9%	Temporal candidates existed, but all candidates from the same buyer had a *valid but incompatible* CPV division ($s_{\text{cpv}} = 0$); sources with a missing CPV are excluded from this label. A diagnostic warning, not a CPV hard gate in the final scoring rule.
Total unlinked	**403**	100%	

The dominant failure mode (`NO_TEMPORAL_PARTNER`, 84.1%) reflects a structural ceiling: buyers who published only one call-for-tender within the study period cannot be linked regardless of algorithm quality. This is the main remaining ceiling on the 36.6% unlinked fraction.

8.3 Linking Rate by Contract Start Year

Figure 4 disaggregates the linking rate by year of contract start. Rates are stable between 55% and 75% for contracts from 2015 to 2022. The sharp drop to 33% in 2023 ($n = 12$) is expected: a contract starting in late 2023 has its estimated renewal window in 2025 or beyond, outside the study period, so most of these contracts are structurally right-censored and cannot be linked regardless of algorithm quality.

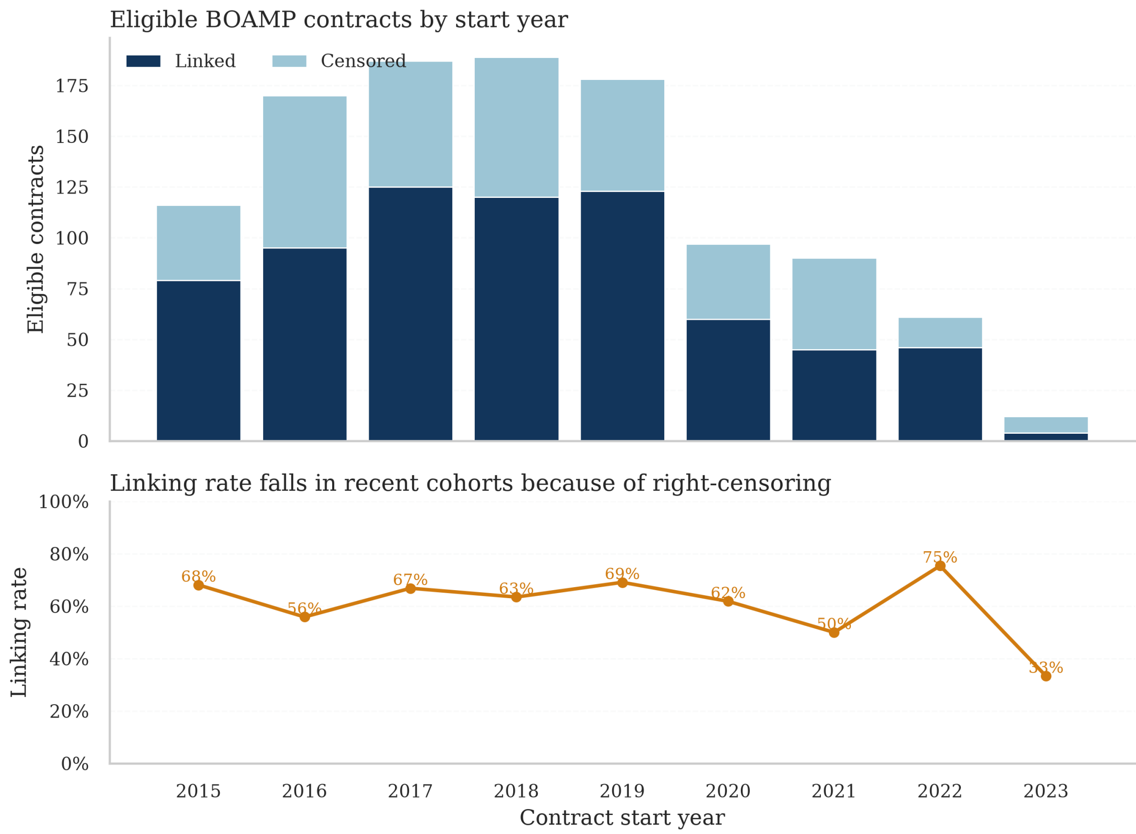


Figure 4: Linked (dark blue) and unlinked (light blue) eligible contracts by year of contract start, with the linking rate overlaid as an orange line (right axis). Rates for 2015–2022 are stable in the range 50–75%. The 2023 cohort ($n = 12$) has a 33% rate due to structural right-censoring: most contracts starting in late 2023 have their expected renewal window after December 2024 and cannot be linked in the current study period.

8.4 Threshold Sensitivity Analysis

The linking rate depends on the minimum text-similarity threshold applied as a hard filter. Figure 5 shows how the number of retained events and the overall eligible linking rate change as the threshold is raised from the current value of 0.20 to 0.50.

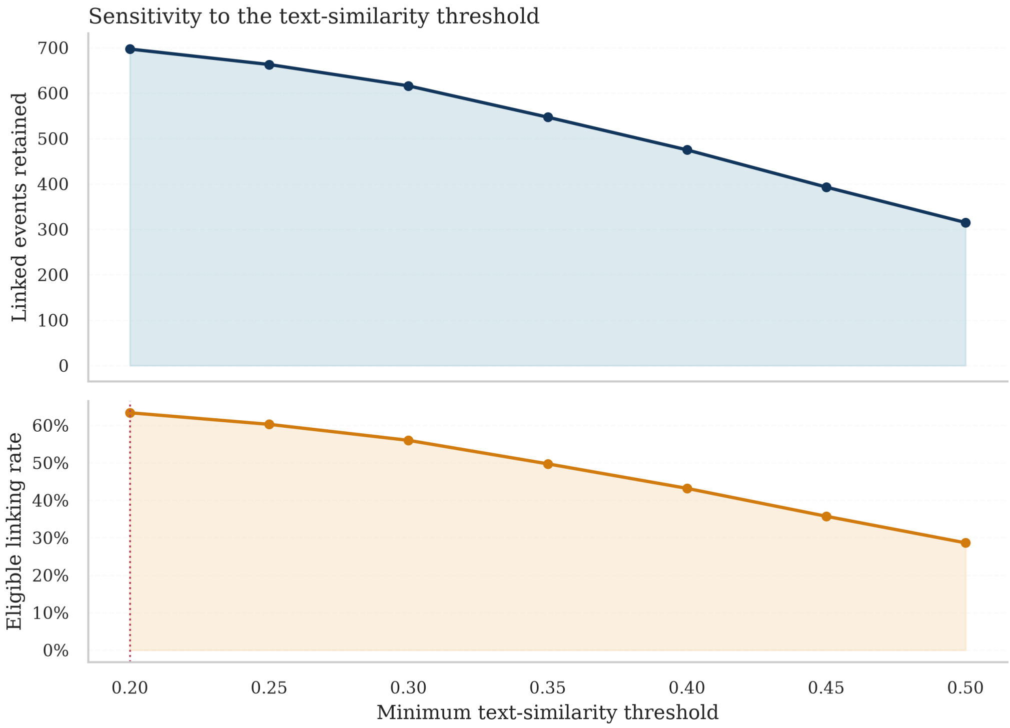


Figure 5: Number of linked events (blue bars, left axis) and eligible linking rate (orange line, right axis) as a function of the minimum text-similarity threshold. At the current threshold of 0.20 (dotted red line), 697 events are retained (63.4%). Raising the threshold to 0.30 reduces the count to 616 (56.0%) and eliminates the 81 pairs with the weakest semantic similarity; raising it to 0.50 yields 315 events (28.6%). Note that this 0.50 threshold is on *text similarity* s_{text} and is distinct from the Model B sensitivity definition in Section 13, which thresholds the *composite* score $S(i, j^*) \geq 0.50$ (494 events). This analysis informs the sensitivity check in Phase 2: the survival model should be run at both 0.20 and 0.30 to assess whether conclusions are stable.

9 Worked Example

Table 20 illustrates the score computation for the first pair in `boamp_renewal_candidates.csv`.

This pair passes both hard filters ($s_{\text{text}} = 0.69 \geq 0.20$; gap $= 33.5 \in [6, 72]$ months) and is selected as the best match for source `15-31464`.

Concrete step

This part illustrates how the renewal-linking score is computed for one concrete candidate pair. The example comes from the first row of `boamp_renewal_candidates.csv`.

The source contract is:

$$\texttt{src} = \texttt{15-31464}$$

Table 20: Score calculation for pair (`15-31464` $\rightarrow$ `17-176560`)

Quantity	Description	Value
t_i	Source start date	2015-03-02
d_i	Declared duration	36 months
$\hat{e}_i$	Estimated end date ($t_i + d_i$)	2018-03-01
t_j	Candidate start date	2017-12-17
$\Delta(i, j)$	Gap ($t_j - t_i$)	33.54 months
s_{text}	ST cosine of *objet* embeddings	0.6922
s_{cpv}	Source CPV missing $\Rightarrow$ excluded (renormalized)	—
s_{temp}	$1 - \lvert 33.54 - 36 \rvert / 12 = 1 - 2.46/12$	0.7951
s_{buyer}	Same buyer group	1.0000
$S(i, j)$	$\dfrac{0.40(0.6922) + 0.20(0.7951) + 0.15(1.0)}{0.75}$	**0.7812**

The candidate renewal notice is:

$$\texttt{cand} = \texttt{17-176560}$$

Both notices belong to the same buyer group:

$$\text{Buyer} = \text{Département de la Mayenne}$$

Source and Candidate Information

Table 21: Source contract and candidate renewal

Element	Notation	Value
Source contract ID	i	`15-31464`
Source start date	t_i	2015-03-02
Declared duration	d_i	36 months
Estimated end date	$\hat{e}_i = t_i + d_i$	2018-03
Candidate contract ID	j	`17-176560`
Candidate start date	t_j	2017-12-17
Buyer match	$b_i = b_j$	Yes

The source contract started on 2015-03-02 and had a declared duration of 36 months. Therefore, the expected renewal date is around March 2018:

$$\hat{e}_i = \text{2015-03-02} + 36 \text{ months} \approx \text{2018-03}$$

The candidate notice was published on 2017-12-17. This is close to the expected end date of the original contract, so it is a plausible renewal candidate.

Step 1: Time Gap

The time gap between the source contract and the candidate is:

$$\Delta(i, j) = t_j - t_i$$

In this example:

$$\Delta(i, j) = \text{2017-12-17} - \text{2015-03-02} = 33.54 \text{ months}$$

The declared duration was:

$$d_i = 36 \text{ months}$$

So the candidate appears:

$$36 - 33.54 = 2.46 \text{ months}$$

before the expected contract end date.

Step 2: Temporal Score

The temporal score is:

$$s_{\text{temp}}(i, j) = 1 - \frac{|\Delta(i, j) - d_i|}{W}$$

with:

$$W = 12$$

Therefore:

$$s_{\text{temp}}(i, j) = 1 - \frac{|33.54 - 36|}{12}$$

$$s_{\text{temp}}(i, j) = 1 - \frac{2.46}{12}$$

$$s_{\text{temp}}(i, j) = 0.7951$$

This is a high temporal score because the candidate appears only about 2.5 months before the expected renewal date.

Step 3: Other Component Scores

The text similarity score is computed using the cosine similarity between the two Sentence-Transformer embeddings:

$$s_{\text{text}}(i, j) = 0.6922$$

This means that the two contract descriptions are semantically close.

The CPV score is:

$$s_{\text{cpv}}(i, j) = \text{NaN (excluded)}$$

because the source CPV code is missing. Instead of inventing a neutral value, the algorithm *excludes* CPV from the composite and renormalizes the remaining component weights. This is important because missing CPV should neither count as evidence against the match nor inject an arbitrary 0.5 signal.

The buyer score is:

$$s_{\text{buyer}}(i, j) = 1.0$$

because the two notices belong to the same buyer group.

Step 4: Composite Score

Because the source CPV is missing, CPV is excluded and the composite is the renormalized weighted sum of the three available components:

$$S(i, j) = \frac{0.40 s_{\text{text}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}}{0.40 + 0.20 + 0.15}$$

Substituting the values:

$$S(i, j) = \frac{0.40(0.6922) + 0.20(0.7951) + 0.15(1.0)}{0.75}$$

$$S(i,j) = \frac{0.2769 + 0.1590 + 0.1500}{0.75} = \frac{0.5859}{0.75}$$

$$S(i,j) = 0.7812$$

So the final composite score is:

$$\boxed{S(i,j) = 0.7812}$$

This matches the `composite_score` reported in the candidate file.

Interpretation

This pair is considered a plausible renewal link because all main signals are consistent:

- the buyer is the same;
- the candidate appears close to the expected contract end date;
- the contract descriptions are semantically similar;
- the CPV code is missing for the source, but this is excluded from the composite (weights renormalized) rather than treated as evidence against the match or scored a neutral 0.5.

The pair also passes the two hard filters:

$$s_{\text{text}} = 0.6922 \geq 0.20$$

and:

$$\Delta(i,j) = 33.54 \in [6, 72]$$

Therefore, this candidate remains in the filtered candidate set. If it has the highest composite score among all surviving candidates for source `15-31464`, it is selected as the best renewal match:

$$j^*(i) = \texttt{17-176560}$$

This example shows exactly how the algorithm transforms raw BOAMP notices into a renewal-linking score. The score is not legal proof of renewal. It is an algorithmic measure of plausibility based on buyer identity, timing, text similarity, and CPV compatibility.

10 Limitations and Risks

L1. No explicit ground truth. BOAMP does not provide a renewal field. The `annonce_lie` signal on ATTRIBUTION notices calibrates text similarity thresholds but is not a direct renewal label. The true precision of the 697 links is unknown without manual validation.

L2. Name-based buyer fragmentation. 87.2% of buyer keys are name-based. Variant spellings of the same public entity (abbreviations, accents, word order) create separate groups, producing structural false negatives that cannot be recovered by tuning thresholds alone.

L3. Duration imputation introduces noise, and declared duration is an administrative ceiling. 25.1% of eligible AO had their duration imputed to 48 months. If the true duration differs significantly, the temporal window is shifted and the correct renewal candidate may fall outside the ±12-month window. More fundamentally, even when duration is not imputed, the declared value is the administrative maximum of the contract (base period plus all renewal tranches), not the expected actual first renewal date. It cannot therefore be used as the survival target. See Section 2.6 for the full analysis.

L4. False positives in sparse-buyer groups. Relaxing CPV from a hard filter to a scored component increases recall but also increases the risk of linking two unrelated contracts from the same buyer that happen to use similar procurement language (e.g. generic IT services descriptions).

L5. Sample scope. The dataset covers only the Pays-de-la-Loire region and is focused on ICT/digital services. Linking rates and score distributions may differ significantly for other regions or sectors.

11 Dataset Reliability Analysis

This section consolidates the quantitative reliability checks performed on `boamp_renewal_links.csv` before it is used as input to the survival model. The goal is to characterise the confidence gradient across the 697 linked pairs and provide a principled basis for the threshold sensitivity analysis recommended in Phase 2.

11.1 Confidence Tiers

Each linked contract is assigned to one of three confidence tiers based on its composite score $S(i, j^*)$:

Table 22: Confidence tiers for linked contracts (event = 1, $n = 697$)

Tier	Criterion	Count	Share
HIGH	$S(i, j^*) \geq 0.70$	122	17.5%
MEDIUM	$0.50 \leq S(i, j^*) < 0.70$	372	53.4%
LOW	$S(i, j^*) < 0.50$	203	29.1%
Censored (event = 0)		403	—

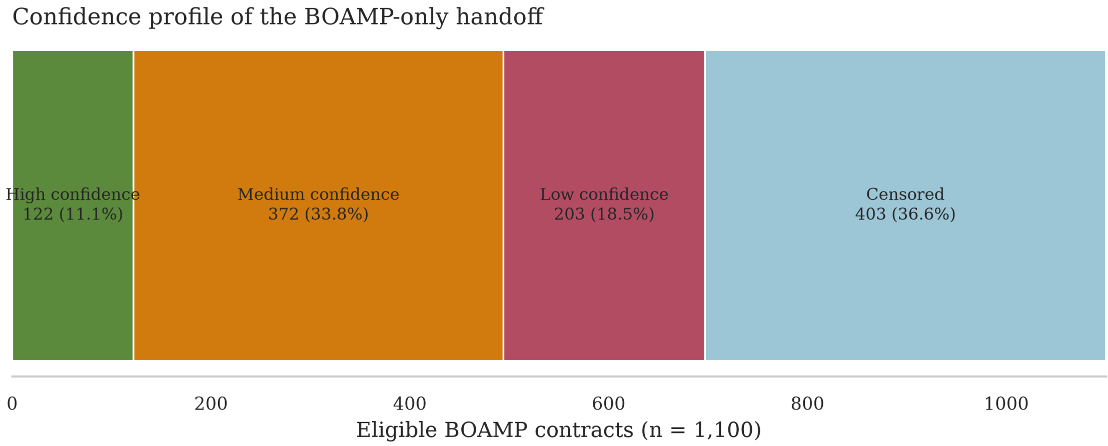


Figure 6: Confidence tier breakdown across all 1,100 eligible contracts. Green = HIGH ($S \geq 0.70$, $n = 122$); orange = MEDIUM ($0.50 \leq S < 0.70$, $n = 372$); red = LOW ($S < 0.50$, $n = 203$); light blue = censored (event = 0, $n = 403$). The HIGH and MEDIUM tiers together cover 494 of the 697 linked contracts (70.9%), providing a reliable core dataset for survival modeling. LOW-tier links represent the greatest false-positive risk and should be downgraded to censored in the sensitivity check.

A finer four-tier classification from the robustness validation further distinguishes *single-candidate* links (only one candidate survived filtering; margin is undefined) from multi-candidate links, and applies a stricter HIGH threshold ($S \geq 0.80$ and margin ≥ 0.05). Figure 7 summarises this breakdown.

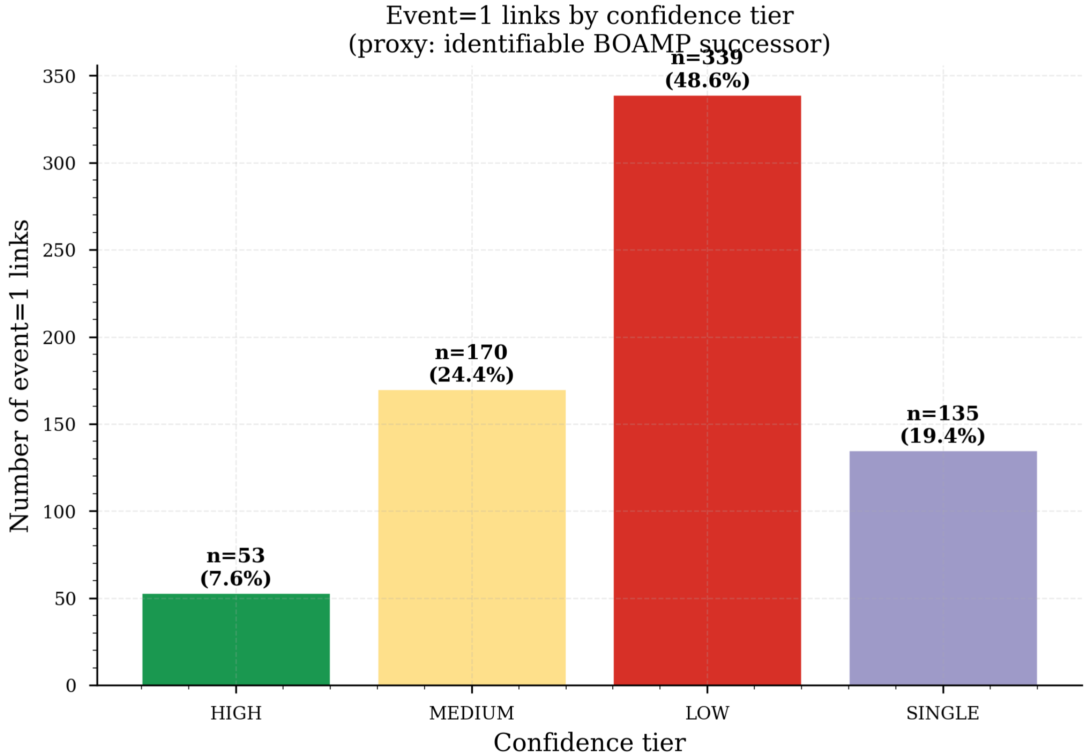


Figure 7: Four-tier confidence classification from the robustness analysis. HIGH: $S \geq 0.80$ and margin ≥ 0.05 ($n = 53$, 7.6% of 697 events); MEDIUM: $0.70 \leq S < 0.80$ or ($S \geq 0.70$ and margin < 0.05) ($n = 170$, 24.4%); LOW: $S < 0.70$ ($n = 339$, 48.6%); SINGLE: single-candidate links with undefined margin ($n = 135$, 19.4%). The `high_confidence_strict` flag used in Model C (Section 11) selects 90 events satisfying $S \geq 0.70$ and margin ≥ 0.05 on multi-candidate links. This covers the entire HIGH tier ($n = 53$, $S \geq 0.80$) plus the subset of MEDIUM-tier links whose score clears 0.70 and whose margin also exceeds 0.05 ($n = 37$). Single-candidate links are recorded separately and are not included in Model C.

For the survival model in Phase 2, the recommended protocol is:

1. **Model A — Primary**: all 697 events (composite ≥ 0.20, text ≥ 0.20).
2. **Model B — Sensitivity check**: 494 events only (drop LOW tier, i.e. composite ≥ 0.50).
3. **Model C — Strict high-confidence**: 90 events only, keeping `event` = 1 only where the absolute score is high *and* the margin over the second-best candidate is clear (`high_confidence_strict`, Section 11.2); all other originally linked contracts are downgraded to censored at study end.
4. If hazard ratios and survival curve shapes are stable across these runs, the lower-confidence links are not distorting conclusions — they add noise but not bias. If results

change substantially, report the sensitivity-check model as the primary.

11.2 Score Disambiguation: Margin Between Best and Second-Best Candidate

For source contracts that had more than one candidate, the margin between the best and second-best composite score measures how clearly the algorithm resolved the ambiguity. A small margin (best ≈ second-best) indicates that an alternative renewal link was nearly as plausible as the selected one.

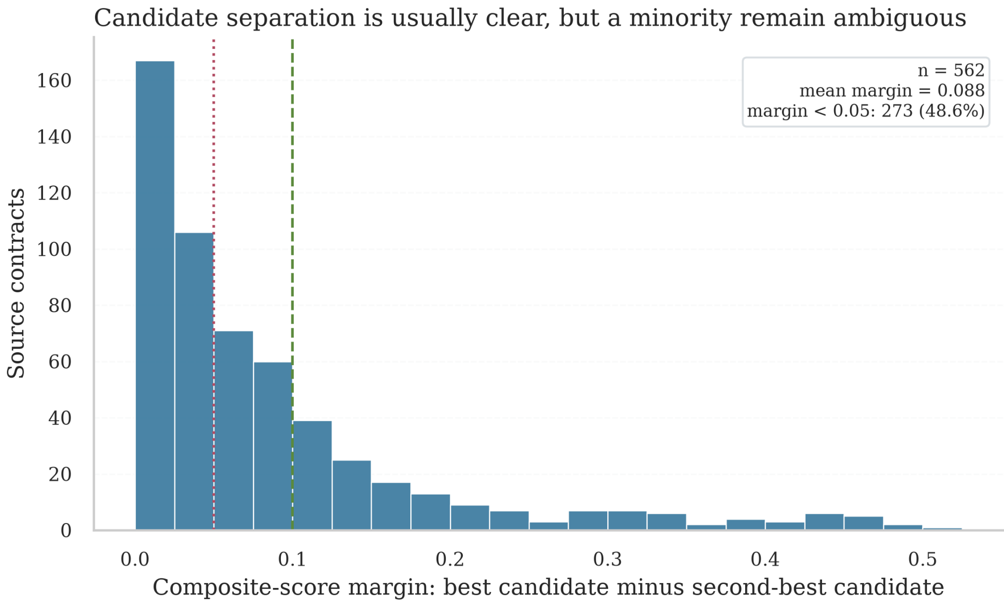


Figure 8: Distribution of the score margin (best composite − second-best composite) for the 562 multi-candidate source contracts. The dotted red line marks the 0.05 threshold below which 48.6% of pairs are considered *ambiguous*: the algorithm selected one candidate, but an alternative was within 0.05 of the same score. The dashed green line at 0.10 marks the *clear winner* zone (28.1% of pairs). These ambiguous pairs are concentrated in the LOW and MEDIUM confidence tiers and are the primary motivation for the manual validation sample recommended in Section 11.

The 48.6% ambiguous-margin rate does not mean the algorithm chose wrongly in those cases — it means the available signals were not strong enough to distinguish the best candidate from the runner-up with high certainty. This is a structural property of the data (multiple contracts from the same buyer in similar time windows) and cannot be resolved by tuning thresholds. Manual validation of a stratified sample is the appropriate mitigation.

Operationalizing the margin: `high_confidence_strict`. We turn this diagnostic into an explicit, conservative flag, written to `boamp_renewal_links.csv`, the Phase 2 handoff, and a dedicated `boamp_link_confidence_diagnostics.csv`. A link is *strict high-confidence* when it satisfies *both* conditions

$$\texttt{event} = 1 \ \wedge \ S(i, j^*) \geq 0.70 \ \wedge \ \Delta \geq 0.05, \qquad \Delta = S(i, j^{(1)}) - S(i, j^{(2)}),$$

i.e. a high absolute score *and* a clear separation from the runner-up. Of the 697 links, **90** qualify. Single-candidate sources ($n = 135$) have an undefined margin and are recorded as `single_candidate_match` rather than auto-promoted to strict, so a "clear winner among several candidates" is never conflated with an "only candidate available". This flag is deliberately conservative: it is *not* a new ground truth but a sensitivity definition used to bound the renewal signal in Model C (Section 11 protocol; results in Table 30).

11.3 Indirect External Calibration (`annonce_lie`)

An indirect calibration is possible using the `annonce_lie` field on ATTRIBUTION notices. These notices back-reference the original call-for-tender, providing external evidence of real procurement activity on those AO records. Among the 466 eligible AO notices that are referenced by at least one ATTRIBUTION notice, 318 were linked (68.2%), compared to 63.4% overall. The algorithm performs *better* on contracts with external evidence of activity, which is the expected direction.

11.4 Summary of Reliability Findings

Table 23: Reliability check results for `boamp_renewal_links.csv`

Check	Result	Interpretation
Internal integrity (duplicates, impossible values)	All clean	No structural data errors
Hard-filter compliance (all event=1 in [6,72] mo)	100% compliant	Algorithm applied correctly
Temporal error mean (observed − declared)	−0.04 months	No systematic temporal bias
Temporal error std	4.87 months	Tight; consistent with ±12 mo window
Duration imputed vs. non-imputed |error|	4.1 vs. 3.7 mo	Negligible difference
Year-by-year rate stability (2015–2022)	50–75%	Stable; 2023 drop is structural
`annonce_lie` calibration	68.2% vs. 63.4%	Positive signal
Low text-similarity links ($s_{\text{text}} < 0.30$)	81 pairs (11.6%)	Highest false-positive risk
Ambiguous score margin (< 0.05)	48.6% of multi-cand.	Structural; not tunable
Strict high-confidence links	90 / 697 (12.9%)	$S \geq 0.70$ & $\Delta \geq 0.05$; Model C core
HIGH + MEDIUM tier share	494 / 697 (70.9%)	Reliable core for modeling

12 Output Files Summary

The notebook produces five output files (four in the `outputs/` directory plus the `boamp_phase2_survival.csv` handoff). The main modeling table is the renewal links file. The other files preserve candidate pairs, aggregate linking statistics, and diagnostic failure patterns for audit and sensitivity analysis.

Table 24: Output files produced by the BOAMP linking workflow

File	Rows	Purpose
`boamp_renewal_links.csv`	1,100	Notebook linking output. One row per eligible AO before final handoff standardisation.
`boamp_phase2_survival.csv`	1,100	**Official modeling input.** BOAMP-only handoff with explicit renewal and censoring duration columns.
`boamp_renewal_candidates.csv`	5,356	All filtered candidate pairs before best-match selection. Useful for threshold sensitivity analysis.
`boamp_linking_stats.csv`	1	Summary statistics: total AO, eligible, linked, linking rate.
`boamp_bias_report.csv`	62	Failure reason × CPV cross-tabulation. Diagnoses which categories are under-linked.

13 Phase 2 — Survival Analysis

The phase numbering in this technical report follows the internal BOAMP pipeline; in the internship guide, the survival-modeling block corresponds to the survival-modeling deliverable.

Phase 2 fits survival models on `boamp_phase2_survival.csv`, the official BOAMP-only handoff produced by Phase 1. The survival time is `observed_duration_months` ($= \Delta(i, j^*)$ for linked contracts; time-to-study-end for censored ones). The event indicator is `event` ($1 =$ renewal linked, $0 =$ right-censored).

13.1 Dataset Overview

Table 25: Phase 2 dataset summary (`boamp_phase2_survival.csv`)

Metric	Value
Total contracts (rows)	1,100
Events (`event = 1`)	697 (63.4%)
Censored (`event = 0`)	403 (36.6%)
Median observed duration (all)	46.8 months
Median renewal gap (events)	37.0 months
Study period	2015-03-02 to 2024-12-29

13.2 Statistical Methods and Notation

Let Y_i denote the Phase 1 algorithmic linking outcome. In Phase 2, the standard survival-analysis event indicator is defined as $\delta_i = Y_i$. Thus $\delta_i = 1$ means that an identifiable BOAMP renewal was linked by the algorithm, and $\delta_i = 0$ means that no renewal was observed under the available data and matching rule.

Let T_i denote the latent time-to-identifiable BOAMP renewal (in months) for contract i and C_i the censoring time (time from contract start to end of study period for right-censored observations). What is observed is the pair $(T_i^{\text{obs}}, \delta_i)$ where:

$$T_i^{\text{obs}} = \min(T_i, C_i), \qquad \delta_i = \mathbf{1}[T_i \leq C_i]. \tag{22}$$

In the output file, T_i^{obs} is stored in `observed_duration_months` and δ_i in `event`.

Kaplan–Meier estimator. Let $t_1 < t_2 < \cdots < t_k$ be the distinct observed event times, d_j the number of events at time t_j, and n_j the number of contracts at risk just before t_j. The product-limit estimator of the survival function is:

$$\hat{S}(t) = \prod_{j:\, t_j \leq t} \left(1 - \frac{d_j}{n_j}\right). \tag{23}$$

Log-rank test. For two strata, the log-rank statistic can be written as:

$$\chi^2_{\text{LR}} = \frac{\left(\sum_j (O_{1j} - E_{1j})\right)^2}{\sum_j V_{1j}}, \tag{24}$$

where O_{1j} is the observed and E_{1j} the expected number of events in stratum 1 at time t_j, and V_{1j} is the hypergeometric variance. For $K > 2$ strata, the test generalises to a vector/matrix form and follows a χ^2 distribution with $K - 1$ degrees of freedom under the null hypothesis of equal survival.

Cox proportional hazards model. The hazard function for contract i with covariate vector $\mathbf{x}_i$ is:

$$h(t \mid \mathbf{x}_i) = h_0(t)\, \exp\!\big(\boldsymbol{\beta}^\top \mathbf{x}_i\big), \tag{25}$$

where $h_0(t)$ is an unspecified baseline hazard. Parameters $\boldsymbol{\beta}$ are estimated by maximising the partial likelihood. For readability, the no-ties form is:

$$\mathcal{L}(\boldsymbol{\beta}) = \prod_{i:\,\delta_i=1} \frac{\exp\!\big(\boldsymbol{\beta}^\top \mathbf{x}_i\big)}{\displaystyle\sum_{j\in\mathcal{R}(T_i^{\mathrm{obs}})} \exp\!\big(\boldsymbol{\beta}^\top \mathbf{x}_j\big)}, \tag{26}$$

where $\mathcal{R}(t) = \{j : T_j^{\mathrm{obs}} \geq t\}$ is the risk set at time t. The hazard ratio for covariate k is $\mathrm{HR}_k = \exp(\beta_k)$; a value > 1 indicates a higher instantaneous renewal rate (faster renewal). Because the empirical durations are measured in months, tied event times can occur; the software implementation handles ties internally. All Cox and AFT models are fitted with a small ridge (L_2) penalty of $\lambda = 0.1$ on the coefficients (the lifelines `penalizer` parameter). This stabilises estimation in the presence of correlated dummy covariates and mildly shrinks the reported hazard ratios and confidence intervals toward the null; exact reproduction of the tabulated values requires the same penalty.

Concordance index (C-index). In-sample ranking ability is measured by Harrell's C-index:

$$C = \frac{\displaystyle\sum_{(i,j):\,T_i^{\mathrm{obs}}<T_j^{\mathrm{obs}},\;\delta_i=1} \mathbf{1}\big[\hat{\eta}_i > \hat{\eta}_j\big]}{\displaystyle\sum_{(i,j):\,T_i^{\mathrm{obs}}<T_j^{\mathrm{obs}},\;\delta_i=1} 1}, \tag{27}$$

where $\hat{\eta}_i = \hat{\boldsymbol{\beta}}^\top \mathbf{x}_i$ is the estimated linear predictor. $C = 0.5$ corresponds to random ranking; $C = 1$ is perfect discrimination.

Accelerated failure time (AFT) models. Parametric AFT models assume:

$$\log T_i = \mu + \boldsymbol{\gamma}^\top \mathbf{x}_i + \sigma \epsilon_i, \tag{28}$$

where the distribution of ϵ_i determines the model family:

$$\epsilon_i \sim \begin{cases} \mathcal{N}(0,1) & \text{log-normal} \\ \text{Logistic}(0,1) & \text{log-logistic} \\ \text{Gumbel}(0,1) & \text{Weibull} \end{cases} \tag{29}$$

Parameters $(\mu, \boldsymbol{\gamma}, \sigma)$ are estimated by maximum likelihood with right-censored observations. Model selection uses the Akaike Information Criterion:

$$\text{AIC} = -2\,\hat{\ell} + 2p, \tag{30}$$

where $\hat{\ell}$ is the maximised log-likelihood and p is the number of free parameters.

Proportional hazards assumption. The PH assumption requires $h(t \mid \mathbf{x}_i)/h(t \mid \mathbf{x}_j)$ to be constant over time. A graphical check uses the log-log transformation: if the PH assumption holds, $\log(-\log \hat{S}_k(t))$ curves across strata k should be approximately parallel.

13.3 Kaplan–Meier Survival Estimates

Global Kaplan–Meier Curve

Figure 9 shows the overall survival function estimated from all 1,100 contracts. The global KM median survival is **48.1 months**, consistent with the dominant 48-month administrative duration cluster in the declared duration distribution. The 95% confidence interval widens beyond 60 months as fewer contracts remain at risk.

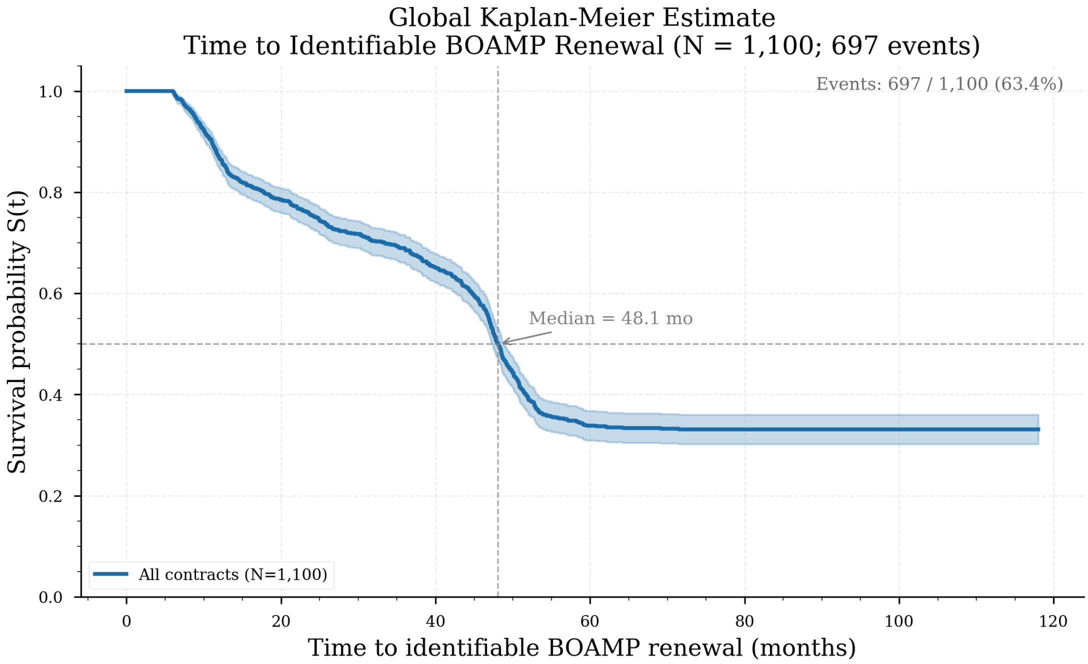


Figure 9: Global Kaplan–Meier survival curve for all 1,100 eligible contracts. The solid blue line is the KM estimate; the shaded band is the 95% pointwise confidence interval. Censored observations are marked by tick marks. Median survival = 48.0 months (vertical dashed line).

Stratified Kaplan–Meier Curves

By service category. Figure 10 stratifies the KM curve by the five most represented service categories (Cybersecurity, IT Services & Consulting, Software & Applications, Telecom & Networks, and Other). The log-rank test gives a test statistic of 6.23 ($p = 0.182$, df $= 4$), indicating no statistically significant difference in renewal timing across ICT categories at the 5% level.

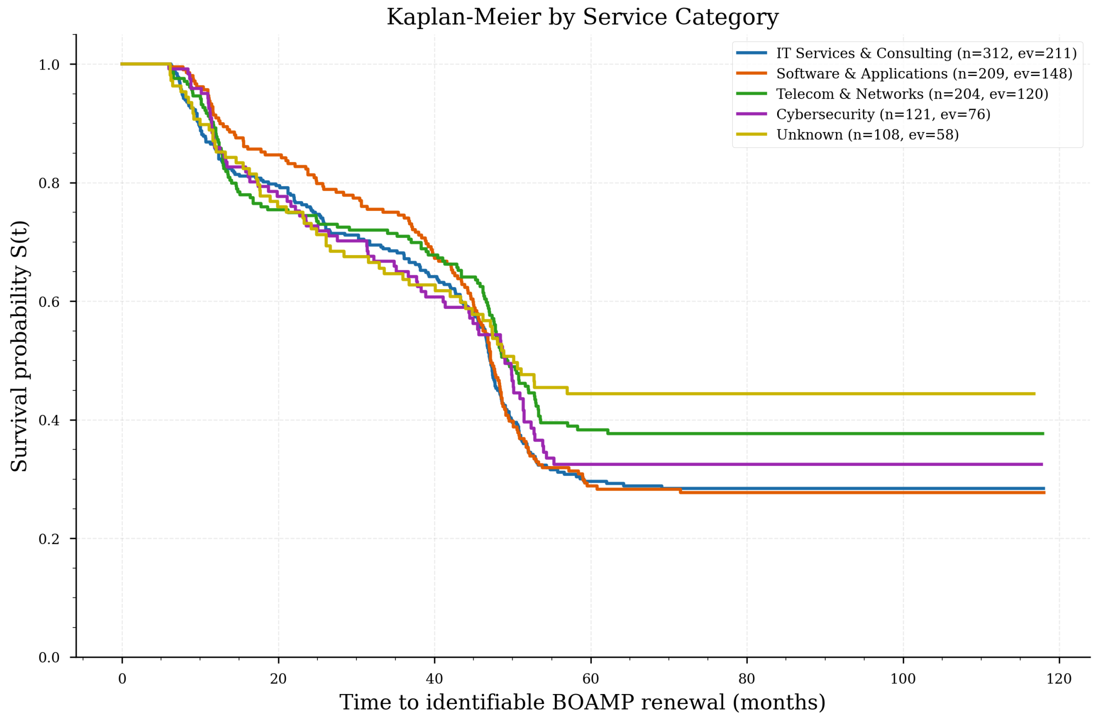


Figure 10: Kaplan–Meier survival curves stratified by service category. Log-rank test: $\chi^2 = 6.23$, $p = 0.182$, df $= 4$. No category shows a statistically significant difference in time-to-renewal.

By declared duration group. Figure 11 stratifies contracts into three duration bins: ≤ 24 months, 25–48 months, and > 48 months. The log-rank test gives $\chi^2 = 62.30$ ($p \approx 0$, df $= 2$), confirming that declared contract duration is the strongest predictor of renewal timing — as expected, since it defines the temporal search window.

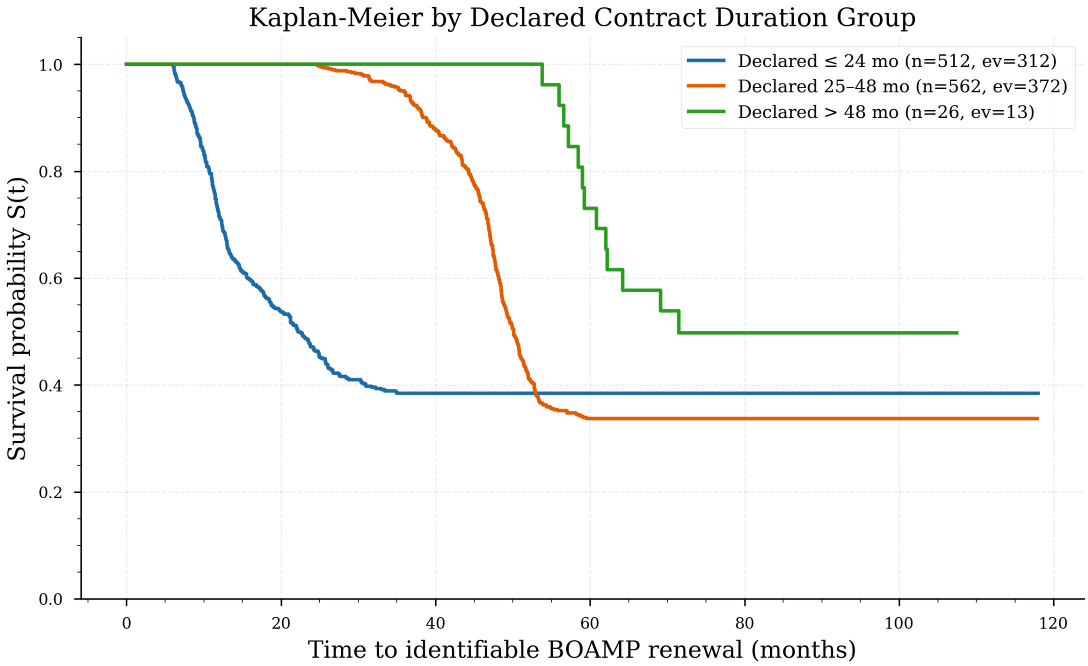


Figure 11: Kaplan–Meier survival curves stratified by declared duration group ($\leq$ 24 mo, 25–48 mo, $>$ 48 mo). Log-rank test: $\chi^2 = 62.30$, $p \approx 0$, df = 2. Shorter declared durations are associated with earlier renewal events.

By imputation status. Figure 12 compares contracts whose duration was observed (non-imputed) against those whose duration was set to the 48-month default (`dur_was_imputed = True`, 25.1% of the dataset). The log-rank test gives $\chi^2 = 24.58$ ($p \approx 0$, df = 1): imputed contracts renew later on average, consistent with the imputed value (48 months) being larger than the median observed duration (24 months among non-imputed AO).

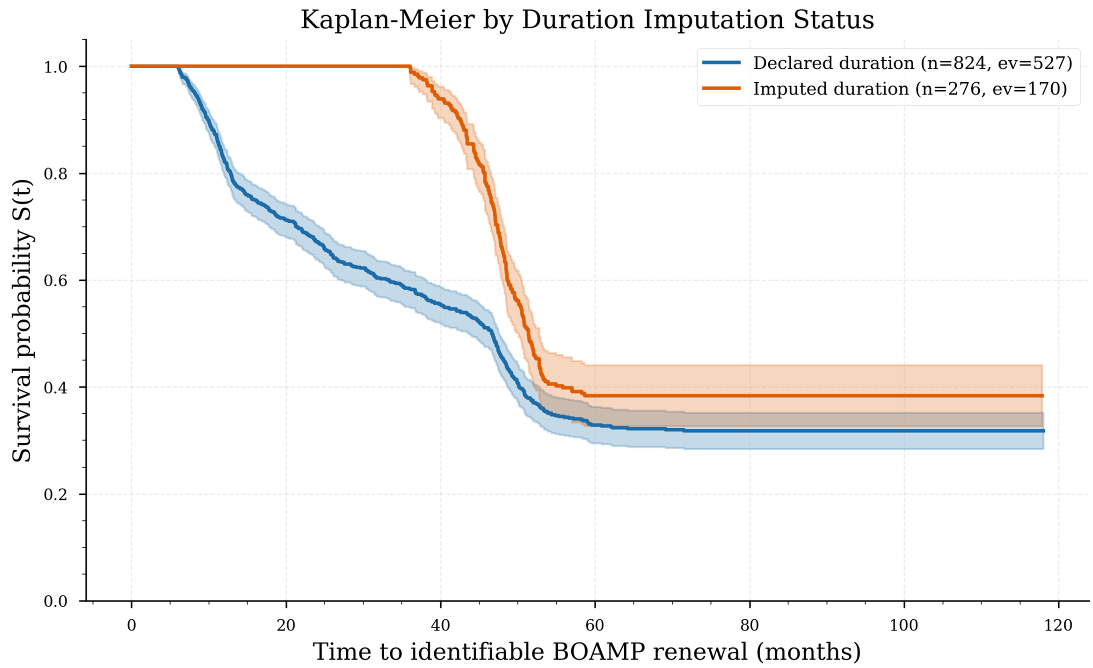


Figure 12: Kaplan–Meier survival curves by duration imputation status. Log-rank test: $\chi^2 = 24.58$, $p \approx 0$, df $= 1$. Contracts with imputed durations (48-month default) renew later than those with observed durations.

13.4 Log-Rank Test Summary

Table 26 collects all log-rank test results. Declared duration group and imputation status are both highly significant. Service category and procedure type are not significant at the 5% level.

Table 26: Log-rank test results for stratified Kaplan–Meier comparisons

Variable	χ^2 statistic	p-value	df
`declared_duration_group` (≤ 24, 25–48, > 48 mo)	62.30	< 0.001	2
`dur_was_imputed` (imputed vs. observed)	24.58	< 0.001	1
`category_label` (5-level ICT category)	6.23	0.182	4
`type_procedure` (3-level collapsed)	2.06	0.357	2

13.5 Cox Proportional Hazards Model

Univariate Cox Results

Table 27 reports univariate Cox models for each covariate. A hazard ratio > 1 indicates a higher instantaneous renewal rate (faster time-to-renewal). Declared duration has the strongest predictive signal (concordance 0.660), followed by imputation status (0.571).

Table 27: Univariate Cox proportional hazards results

Covariate	HR	95% CI	p-value	Concordance
`declared_duration_months`	0.985	[0.981, 0.989]	< 0.001	0.660
`dur_was_imputed`	0.653	[0.549, 0.777]	< 0.001	0.571
`start_year`	1.166	[1.118, 1.216]	< 0.001	0.590
— *Category (ref: Cybersecurity)* —				
IT Services & Consulting	1.158	[0.985, 1.362]	0.075	0.513
Other	0.847	[0.704, 1.018]	0.077	0.505
Software & Applications	1.092	[0.911, 1.309]	0.343	0.499
Telecom & Networks	0.887	[0.728, 1.080]	0.233	0.507
— *Procedure (ref: Négocié)* —				
OUVERT	1.114	[0.959, 1.294]	0.159	0.526
Other procedure	0.664	[0.391, 1.128]	0.130	0.504
PROCEDURE_ADAPTE	0.909	[0.778, 1.061]	0.227	0.522

Multivariate Cox Model

Table 28 reports the final multivariate Cox model including all covariates. Two predictors are statistically significant at the 1% level: `declared_duration_months` and `start_year`. The model is fitted on the 1,097 contracts (696 events) with complete category and procedure covariates; three rows are dropped for missing covariate values. The model concordance (C-index) is **0.6524**, indicating moderate in-sample ranking ability beyond a random ranking.

Table 28: Multivariate Cox proportional hazards model (C-index = 0.6524)

Covariate	HR	95% CI	*p*-value
`declared_duration_months`	0.991	[0.987, 0.995]	$< 0.001^{***}$
`start_year`	1.094	[1.050, 1.140]	$< 0.001^{***}$
`dur_was_imputed`	0.864	[0.718, 1.040]	0.123
— *Category (ref: Cybersecurity)* —			
IT Services & Consulting	1.124	[0.923, 1.367]	0.244
Software & Applications	1.180	[0.951, 1.463]	0.132
Telecom & Networks	0.911	[0.730, 1.137]	0.409
Other	0.869	[0.704, 1.072]	0.191
— *Procedure (ref: Négocié)* —			
OUVERT	0.979	[0.789, 1.216]	0.850
PROCEDURE_ADAPTE	1.027	[0.820, 1.286]	0.818
Other procedure	0.660	[0.402, 1.084]	0.101
*** $p < 0.001$			

Figure 13 shows the forest plot of hazard ratios with 95% confidence intervals for all covariates in the multivariate model.

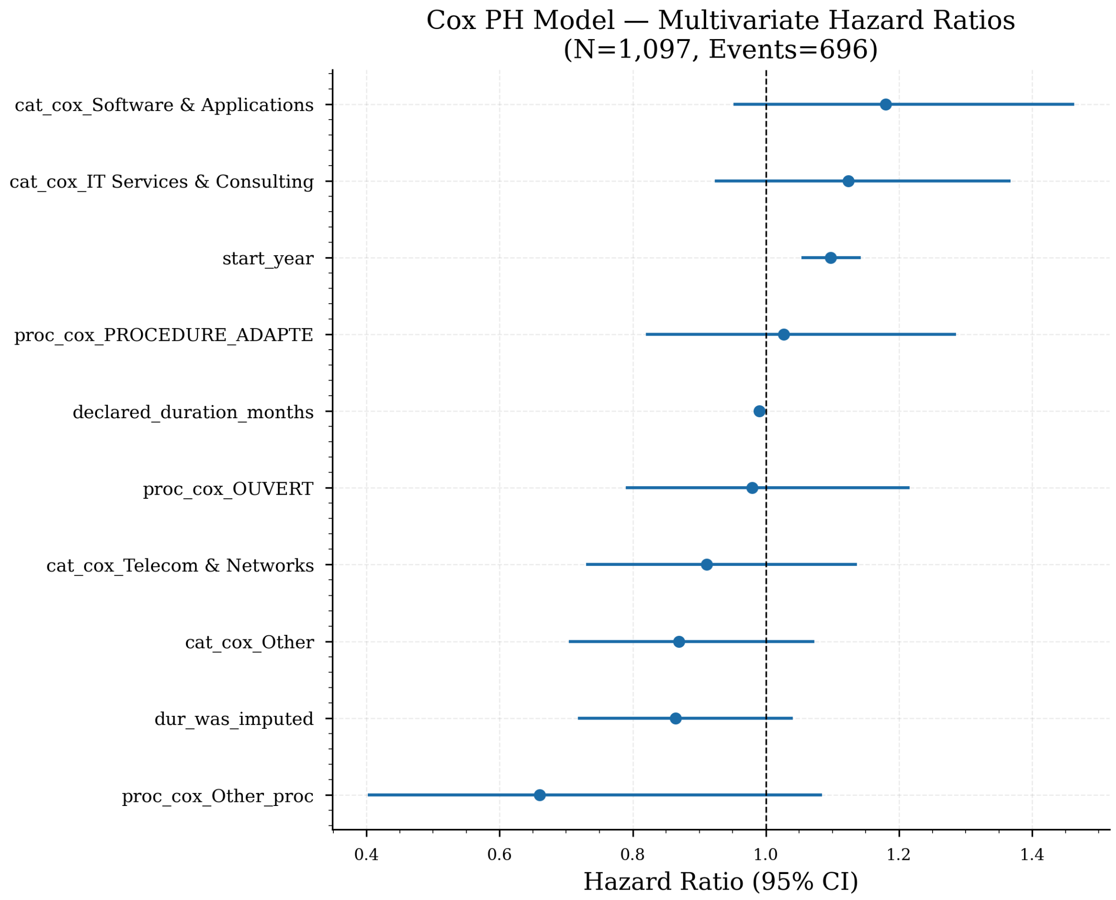


Figure 13: Forest plot of multivariate Cox hazard ratios. Points indicate the estimated HR; horizontal bars are 95% confidence intervals. The vertical dashed line at HR = 1 indicates no effect. Blue/filled markers are statistically significant at the 5% level.

Proportional Hazards Assumption Check

The proportional hazards assumption was checked two ways: a formal Schoenfeld residual test (lifelines `check_assumptions`) and a graphical log-log transformation of the KM survival function. The Schoenfeld test flags a violation for `declared_duration_months` ($p < 5\times10^{-5}$), the strongest predictor in the model; the remaining covariates pass. This indicates a time-varying (non-proportional) effect for declared duration. We interpret this with caution rather than as invalidating: the Schoenfeld test is highly sensitive to functional-form misspecification, and the likely cause here is a mildly non-linear relationship between declared duration and the log-hazard rather than a true reversal of effect over time. Two facts limit the practical impact: (i) the parametric AFT models reported below do not rely on the PH assumption and yield the same directional conclusions; and (ii) the `declared_duration_months` hazard ratio (HR = 0.991) is consistent in sign and magnitude across the univariate, multivariate, and sensitivity specifications. A fully rigorous treatment would stratify on binned declared duration or add a time interaction; this is flagged as future work.

Figure 14 shows the $\log(-\log(\hat{S}(t)))$ curves stratified by imputation status. Approximately parallel lines are consistent with the PH assumption for that covariate; crossing or diverging lines suggest a violation.

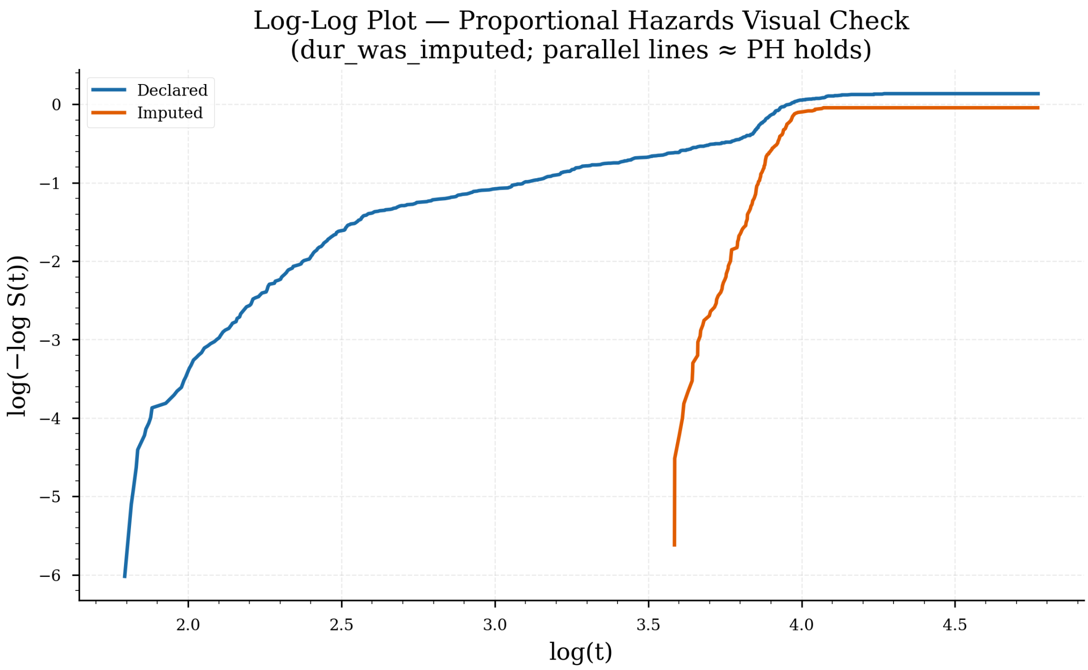


Figure 14: Log-log plot for the proportional hazards assumption check, stratified by duration imputation status (declared vs. imputed). The curves are approximately parallel over the observable time range, consistent with the PH assumption for this covariate. Note that the formal Schoenfeld test flags `declared_duration_months` as non-proportional (see text).

13.6 Parametric Survival Models

Three accelerated failure time (AFT) models were fitted to assess parametric alternatives to the semi-parametric Cox model. Table 29 compares goodness-of-fit (AIC) and concordance across distributional families.

Table 29: Parametric AFT model comparison

Model	AIC	Concordance
Log-Normal	**7 119.8**	**0.672**
Log-Logistic	7 194.3	0.677
Weibull	7 320.3	0.636

The log-normal model achieves the lowest AIC (7 119.8) and a concordance of 0.672, marginally above the Cox model (0.654). The log-logistic model achieves the highest concordance (0.677) at the cost of a higher AIC. The Weibull model performs worst on both metrics. These results suggest that the renewal gap distribution has a heavier tail than the Weibull's exponential hazard allows, and that the non-monotone hazard shape of the log-normal/log-logistic models better captures the data.

Figure 15 overlays the three parametric predicted survival curves against the empirical Kaplan–Meier estimate.

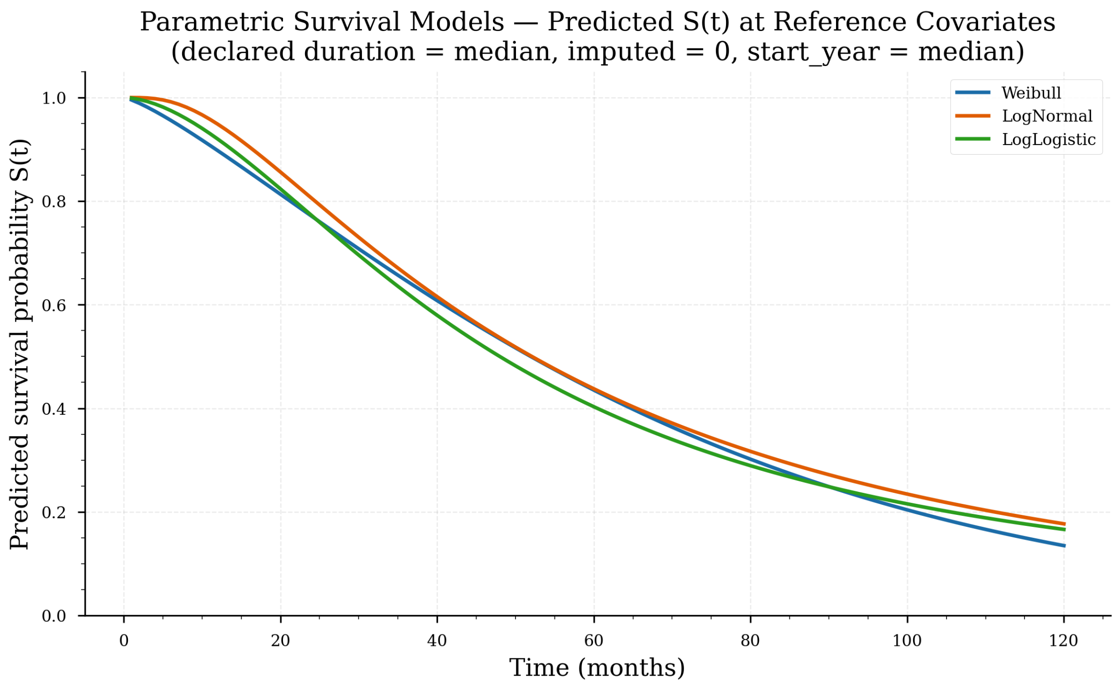


Figure 15: Parametric survival predictions from the three AFT models (Weibull, Log-Normal, Log-Logistic) overlaid on the empirical Kaplan–Meier curve. The log-normal fit is closest to the KM estimate, consistent with its lower AIC. The Weibull model underestimates survival at early times and overestimates at late times.

13.7 Sensitivity Analysis: Model A vs. Model B vs. Model C

Three model configurations were compared to assess the impact of low-confidence links (Section 11) on the survival estimates. Model B drops the LOW tier (composite < 0.50); Model C is the strict margin-based definition of Section 11.2, which retains only links that are both high-scoring and clearly separated from the runner-up.

Table 30: Sensitivity analysis: Model A (baseline) vs. Model B (conservative) vs. Model C (strict high-confidence)

Metric	Model A	Model B	Model C
Event criterion	text $\geq$ 0.20 (all accepted)	composite $\geq$ 0.50	$S \geq 0.70$ & $\Delta \geq 0.05$
Events	697	494	90
Censored	403	606	1 010
Event rate	63.4%	44.9%	8.2%
KM median survival	48.1 mo	52.7 mo	not reached
Cox C-index	0.6524	0.6436	0.6295

Interpretation. Restricting to high- and medium-confidence links (Model B) shifts the survival curve upward, with the KM median rising from 48.1 to 52.7 months, consistent with the excluded LOW-tier links being concentrated at shorter observed gaps (lower text similarity often corresponds to borderline temporal matches). Model C is far more aggressive — by censoring all but the 90 strict links, the event rate collapses to 8.2% and the KM median is no longer reached, so the apparent *level* of renewal risk drops sharply, exactly as expected when most proxy events are removed. The absolute event rate and KM survival level are therefore sensitive to the event definition. The decisive observation is that the Cox concordance remains stable across all three definitions (0.652, 0.644, 0.630): the smaller event counts reduce power but the *ranking* of renewal risk by covariate is robust. The qualitative conclusions — declared duration and start year as the dominant predictors, no significant category or procedure effect — are stable across all three models. Model C is a conservative lower bound, not a corrected estimate of the true event rate.

Figure 16 overlays the KM curves from the three models.

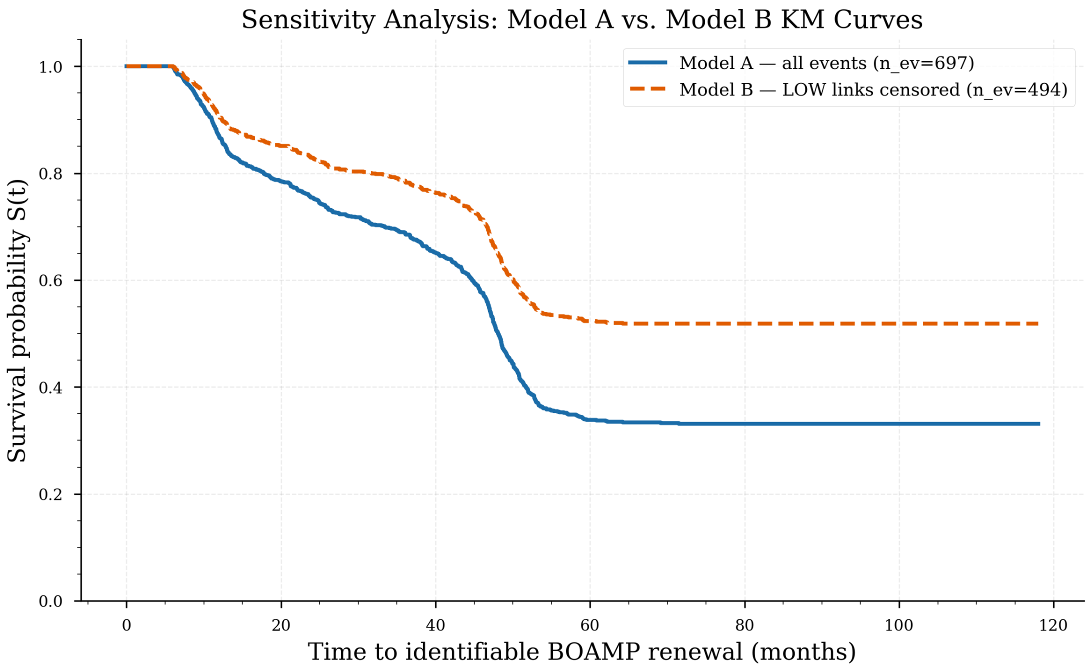


Figure 16: Kaplan–Meier comparison: Model A (all 697 linked contracts, median 48.1 mo) vs. Model B (494 high/medium-confidence contracts, median 52.7 mo) vs. Model C (90 strict high-confidence links, median not reached). Models A and B share broadly consistent curve shapes; Model C sits far higher (most contracts censored), so its survival curve never reaches 0.5. The stable Cox concordance across all three (0.652, 0.644, 0.630) confirms that lower-confidence links add noise but do not appear to introduce a dominant systematic bias in the ranking of renewal risk.

13.8 Extended Robustness and Validation Checks

The three-model sensitivity comparison (Models A/B/C) in the preceding subsection tests the event definition at three levels. This subsection presents five additional robustness checks derived from the dedicated validation analysis (`validation_robustness/`). They address, respectively: the full threshold sweep beyond just ≥ 0.50; the effect of removing uncertain links entirely; the quality of the individual score components; the discriminative ability of the scoring function against null comparators; and the covariate balance between linked and censored contracts.

1. Full Threshold Sensitivity (7 Scenarios)

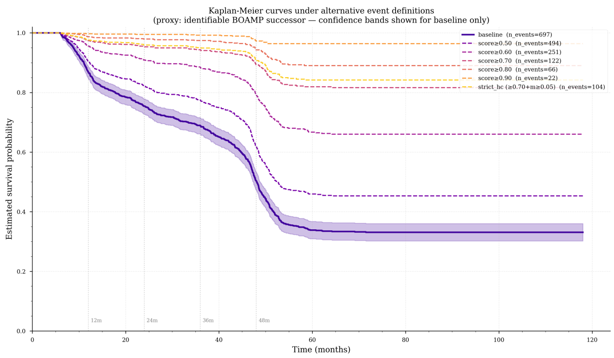
Table 31 extends the sweep across seven scenarios: from the baseline (all accepted links, no composite floor) up to score ≥ 0.90, plus the strict high-confidence variant.

Table 31: Full threshold sensitivity: survival model outcomes under increasingly strict event definitions. KM median in months; "∞" = median never reached within study window. HR values are from the multivariate Cox model.

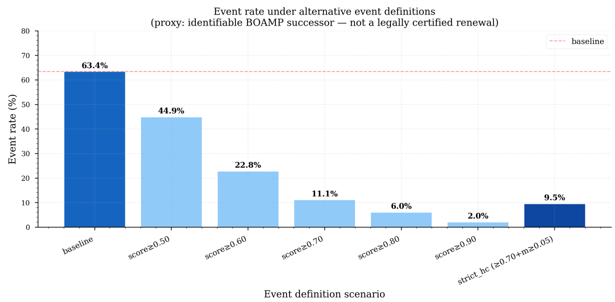
Scenario	**events**	**rate%**	**KM med.**	**Cox C**	**HR dur.**	**HR yr**
Baseline (all accepted links)†	697	63.4	48.1	0.6524	0.992	1.094
Score ≥ 0.50	494	44.9	52.7	0.6436	0.992	1.074
Score ≥ 0.60	251	22.8	∞	0.6221	0.995	1.068
Score ≥ 0.70	122	11.1	∞	0.6317	0.995	1.035
Score ≥ 0.80	66	6.0	∞	0.5935	0.999	1.015
Score ≥ 0.90	22	2.0	∞	0.5558	1.001	1.013
Strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$)	104	9.5	∞	0.6375	0.995	1.046

† Baseline retains all accepted links (`text_similarity` ≥ 0.20 hard filter only; no composite floor). The observed minimum composite score for accepted links is approximately 0.37, arising from the weighting formula applied to passing pairs.

Key observations. (i) The Cox C-index stays above 0.59 across all thresholds, including the extreme ≥ 0.90 case (22 events). (ii) The direction of both significant hazard ratios (HR < 1 for declared duration, HR > 1 for start year) is preserved at every threshold, confirming directional stability. (iii) The KM median becomes non-identifiable once the event rate falls below $\approx$23%, because the study window is too short to observe the median survival under such sparse events. (iv) The strict H.C. variant (104 events including single-candidate links with $S \geq 0.70$) gives C $= 0.638$ — higher than the strict_hc_only variant (90 events, C $=$ 0.630) that excludes single-candidate links.



(a) KM survival curves at each threshold. Tighter thresholds shift the curve upward (fewer events, slower apparent renewal). The baseline and ≥ 0.50 curves nearly overlap, indicating the LOW-tier links do not strongly distort the level.



(b) Event rate and Cox C-index as a function of score threshold. The C-index plateau from the baseline through score ≥ 0.70 (range 0.622–0.652) confirms that adding lower-confidence links does not degrade ranking ability.

Figure 17: Full threshold sensitivity: survival model robustness across 7 event definitions.

2. Uncertain-Link Exclusion Analysis

Rather than downgrading uncertain links to censored (as in Model C), an alternative strategy is to *remove* them from the denominator entirely, reducing the analysis sample to contracts with either a clearly linked event or a definitively unlinked censoring.

Table 32: Uncertain-link exclusion: three analysis variants. `exclude_uncertain` removes all contracts with score < 0.50 or margin < 0.05 (reducing N from 1,100 to 763). `strict_hc_only` keeps all 1,100 contracts but censors all but the 90 strict high-confidence events.

Variant	N	events	rate%	KM med.	Cox C
Baseline	1,100	697	63.4	48.1 mo	0.6524
Exclude uncertain	763	360	47.2	59.0 mo	0.6014
Strict H.C. only	1,100	90	8.2	∞	0.6295

Excluding uncertain links shifts the KM median upward to 59.0 months and the C-index to 0.601, still above the 0.5 random-chance floor. The qualitative conclusion — declared duration and start year as dominant predictors — persists. The drop in C-index relative to the baseline (0.652 → 0.601) reflects both the reduced sample size and the removal of the higher-variance LOW-tier links that, despite their uncertainty, still carry discriminative signal for ranking.

3. Score Component Diagnostics

Table 33 shows the distribution of each scoring component across the 697 accepted links. The temporal score is the highest-mean component; the CPV match score is the lowest, reflecting frequent reclassification across procurement cycles. The score margin median (0.052) is just above the 0.05 ambiguity threshold, meaning roughly half of multi-candidate links are near the boundary.

Table 33: Score component distributions for the 697 event-1 links. CPV count (646) is lower because 51 links have a missing CPV on at least one side. Score margin is defined only for 562 multi-candidate links.

Component	n	Mean	Std	Q1	Median	Q3
Composite score	697	0.579	0.141	0.489	0.555	0.653
Text similarity	697	0.517	0.200	0.368	0.475	0.619
CPV match score	646	0.301	0.366	0.000	0.200	0.400
Temporal score	697	0.706	0.253	0.587	0.775	0.914
Score margin (Δ)	562	0.088	0.105	0.019	0.052	0.108

Figure 18 checks whether composite score correlates with declared contract duration, which would indicate that short-duration contracts are systematically harder to link (a selection bias concern). No strong correlation is observed, supporting the use of score-based thresholds independently of contract duration.

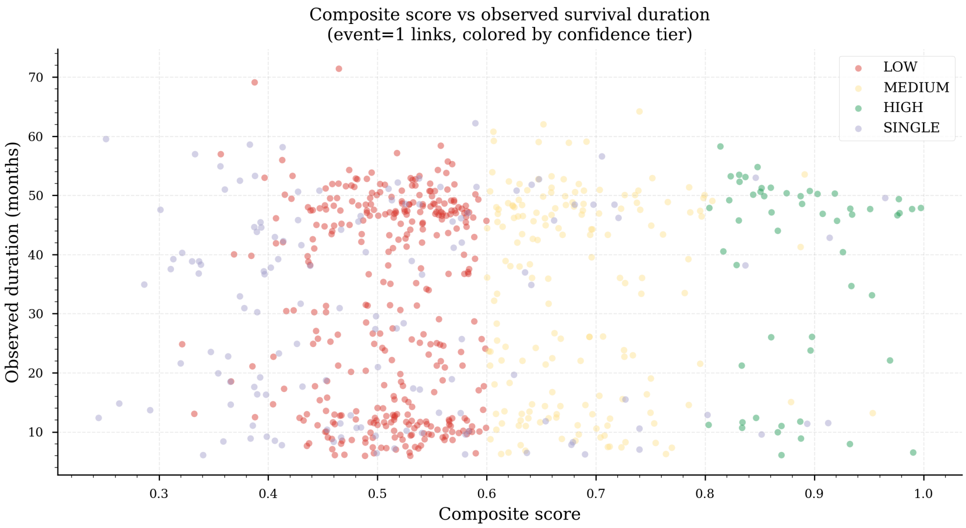


Figure 18: Composite score vs. declared contract duration (months) for the 697 accepted links. No strong duration-linked selection bias is apparent; short contracts (12 months) and long contracts (48+ months) achieve similar score distributions.

4. Placebo / Negative-Control Check

To verify that the scoring function discriminates rather than assigning high scores uniformly, three groups are compared: the winning link (real event), the runner-up (the second-best candidate, not accepted), and a synthetic "no-buyer" candidate constructed by stripping the buyer-identity signal.

Table 34: Placebo check: composite score distributions for real winning links, runner-up candidates, and synthetic no-buyer candidates. $\% \geq 0.70$ = fraction reaching the HIGH confidence threshold.

Group	n	**Mean**	**Median**	**%** ≥ 0.70
Real winning links (event = 1)	697	0.579	0.555	17.5
Runner-up (rejected candidate)	4,659	0.435	0.431	0.5
Synthetic no-buyer	697	0.425	0.399	6.2

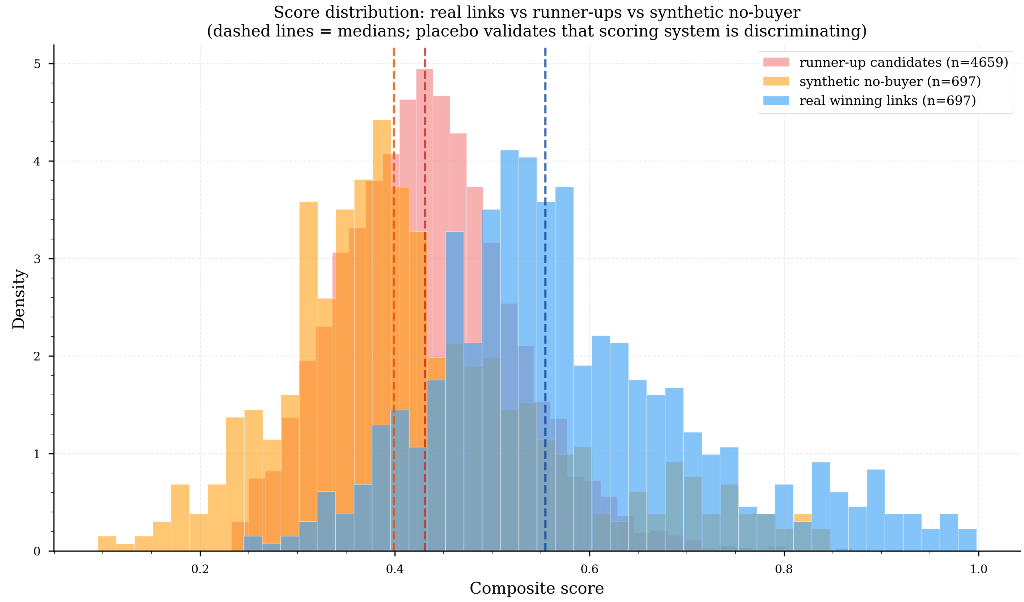


Figure 19: Score distribution for real winning links (green), runner-up candidates (orange), and synthetic no-buyer candidates (red). The clear separation — winner mean 0.579 vs. runner-up 0.435 — confirms that the composite score discriminates meaningfully. Only 0.5% of runner-ups exceed the 0.70 HIGH threshold, compared to 17.5% of winners.

The gap between winners (0.579) and runner-ups (0.435) represents the net discriminative contribution of the four scoring components. The synthetic no-buyer group (0.425) is slightly below runner-ups because some runner-up candidates share the same buyer and thus retain the buyer-identity signal; the small difference (0.435 vs. 0.425) confirms the buyer component contributes positively but is not the dominant signal (the text and temporal components explain most of the winner/runner-up gap).

5. Event / Censoring Bias Check

An important validity concern is whether the contracts that received `event = 1` differ systematically from censored contracts on the covariates used in the survival model. A systematic difference could introduce bias into the Cox hazard ratios if the censoring is informative.

Table 35: Covariate balance between event-1 (linked) and event-0 (censored) contracts. SMD = standardised mean difference; Δ = difference in proportions. SMD < 0.20 is conventionally considered small.

Variable	**event = 1 (n = 697)**	**event = 0 (n = 403)**	**SMD / Δ**
Declared duration (months)	32.3 ± 17.0	30.0 ± 18.6	0.128
Contract start year	2018.0 ± 2.0	2018.1 ± 2.1	−0.029
Duration imputed	24.4%	26.3%	−0.019
Amount non-missing	16.6%	26.6%	−0.099
CPV code present	95.8%	94.3%	0.015
Buyer identified by SIRET	0.0%	0.0%	0.000
N candidates considered	7.7 ± 8.0	0.0 ± 0.0	1.362 (structural)

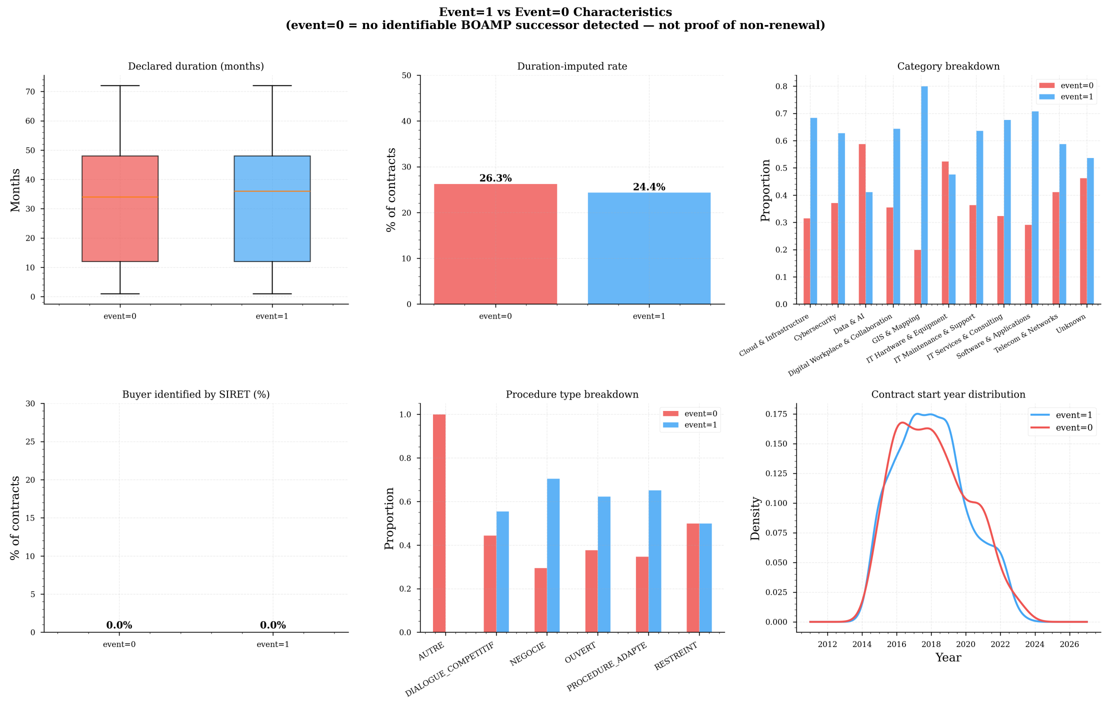


Figure 20: Covariate balance plots comparing event-1 and event-0 contracts. Bars show the standardised mean difference (numeric variables) or difference in proportions (binary variables). All modeling covariates have SMD < 0.15, indicating good balance. The large SMD for N_candidates (1.36) is structural and expected: censored contracts have no temporal partner by definition, so zero candidates were evaluated.

Interpretation. All covariates used in the survival model (declared duration, start year, imputation flag) have SMD below 0.15, indicating that the linked and censored groups are well balanced on these dimensions. The one notable imbalance is amount non-missing

($\Delta = -0.099$): censored contracts are more likely to have a reported amount, suggesting that amount-bearing contracts are slightly harder to re-link — possibly because high-value contracts are more likely to be renegotiated without a new BOAMP call. This does not affect the survival model (amount is not a covariate) but is worth monitoring in future analyses. The dominant source of censoring is structural right-censoring of the 2022–2024 cohort, whose contracts' expected renewal windows fall after the 2024-12-31 study close.

13.9 Summary of Phase 2 Findings

F1. Global median renewal time: 48 months. The KM estimate confirms that the 48-month administrative duration is the dominant pattern in public procurement renewal. The distribution is right-skewed: roughly 25% of contracts renew within 24 months.

F2. Two statistically significant predictors. In the multivariate Cox model, only `declared_duration_months` (HR = 0.991, $p < 0.001$) and `start_year` (HR = 1.094, $p < 0.001$) are significant. Shorter declared durations predict faster renewal. The positive `start_year` hazard ratio should be interpreted cautiously: it may reflect calendar-period effects, publication practices, and study-window truncation (2021–2024 contracts are largely right-censored) rather than a genuine acceleration of procurement renewal cycles.

F3. No significant category or procedure effect. None of the ICT service categories or procurement procedure types differ significantly from their reference in the multivariate model ($p > 0.10$ for all). Log-rank tests confirm this at the bivariate level.

F4. Log-normal AFT model preferred. Among parametric models, the log-normal achieves the best trade-off between fit (AIC = 7 119.8) and concordance (0.672), outperforming the Cox model marginally. The Weibull model is the worst fit.

F5. Sensitivity check supports qualitative stability. Restricting to medium- and high-confidence links (Model B) shifts the KM median from 48.1 to 52.7 months, while leaving covariate effects qualitatively unchanged. Model C is far more aggressive: censoring all but the 90 strict high-confidence links collapses the event rate to 8.2% and the KM median is no longer reached. The primary model (Model A) is retained as the official result.

13.10 Operational 12/24-Month Renewal Risk Prediction

Business purpose. The preceding survival analysis estimates the *historical* timing of renewals: how long, on average, a digital-sector public contract takes before an identifiable BOAMP renewal signal appears. This subsection translates the fitted survival model into *operational* risk indicators for Gigalis. For each contract still active at a prediction date, it estimates the probability that the contract generates an identifiable BOAMP renewal signal within the next 12 or 24 months, and aggregates these probabilities into buyer-level and segment-level rankings to prioritise renewal-campaign effort.

Prediction model. The **log-normal AFT** model of Section 13 is used as the main prediction model, because it has the best (lowest) AIC among the parametric survival models (AIC = 7 119.8; Table 29). It is applied with the same covariates and design matrix as the survival analysis. The prediction date is the study end date, **2024-12-31**; the contract age at that date is $a_i = (2024\text{-}12\text{-}31 - \texttt{start\_date}_i)$ expressed in months. Predictions are produced for the 1,097 contracts in the model design matrix (the 3 contracts with missing `type_procedure` are excluded, as in the Cox/AFT models).

Conditional renewal probability. For a contract i with current age a_i, covariates X_i, and prediction horizon $h \in \{12, 24\}$ months, the conditional probability of an identifiable renewal within the horizon, given that the contract has not yet shown one, is

$$\hat{p}_{i,h} = \Pr(T_i \le a_i + h \mid T_i > a_i, X_i) = \frac{\hat{S}(a_i \mid X_i) - \hat{S}(a_i + h \mid X_i)}{\hat{S}(a_i \mid X_i)},$$

where T_i is the time to an identifiable BOAMP renewal, a_i is the contract age at the prediction date, h is the prediction horizon (12 or 24 months), and $\hat{S}(t \mid X_i)$ is the survival probability predicted by the selected log-normal AFT model. By construction $\hat{p}_{i,24} \ge \hat{p}_{i,12}$ (the survival function is monotone non-increasing); probabilities are clipped to $[0, 1]$ and the 0/0 case ($\hat{S}(a_i) \approx 0$) is set to zero. Across the 1,097 contracts the 12-month probability ranges from 0.075 to 0.435 (mean 0.190) and the 24-month probability from 0.145 to 0.659 (mean 0.332); the expected number of identifiable renewals is 208 within 12 months and 365 within 24 months.

Risk tiers. Contracts are assigned an operational risk tier from $\hat{p}_{i,12}$: **High** (≥ 0.50), **Medium** ($0.25 \le p < 0.50$), and **Low** ($p < 0.25$). On the current data this yields 0 High, 222 Medium, and 875 Low contracts: because most contracts are already several years old, no contract reaches a 12-month conditional probability of 0.50. These are *operational* thresholds, to be calibrated later with the supervisor and with the manual-validation sample.

Output files. Three operational tables are produced (Table 36).

Table 36: Operational prediction output files (in `reports/tables/survival/`)

File	Rows	Purpose / main columns
`renewal_risk_12_24_months.csv`	1,097	Contract-level predictions: one row per scored contract. Main columns: `contract_id`, `buyer_key`, `category_label`, `age_months`, `p_renewal_12m`, `p_renewal_24m`, `risk_tier`, `model_used`.
`buyer_renewal_risk_ranking.csv`	306	Buyer-level ranking by expected renewals. Main columns: `buyer_key`, `n_contracts`, `expected_renewals_12m/_24m`, `max/mean_p_renewal_12m`, `high_risk_contracts_count`.
`segment_renewal_risk_ranking.csv`	11	Segment-level ranking by expected renewals. Main columns: `category_label`, `n_contracts`, `expected_renewals_12m/_24m`, `mean_p_renewal_12m/_24m`, `high_risk_contracts_count`.

Top contracts. Table 37 lists the 10 contracts with the highest 12-month renewal-risk probability.

Table 37: Top 10 contracts by $\hat{p}_{i,12}$ (12-month renewal-risk probability)

Contract ID	Buyer (normalised key)	Segment	p_{12}	p_{24}
BOAMP:23-98097	vendee grand littoral	Software & Applications	0.435	0.659
BOAMP:22-166149	gie des epl de vendee	Software & Applications	0.425	0.647
BOAMP:23-137061	clisson	Telecom & Networks	0.402	0.626
BOAMP:23-30054	maine et loire	Cybersecurity	0.396	0.615
BOAMP:22-94809	com d agglo reg nazairienne est.	Software & Applications	0.385	0.603
BOAMP:22-153148	des pays de la loire	IT Services & Consulting	0.384	0.601
BOAMP:22-112462	semitan	IT Services & Consulting	0.383	0.600
BOAMP:23-56091	ubs	Telecom & Networks	0.383	0.601
BOAMP:23-31948	soc publique reg abe fontevraud	Telecom & Networks	0.379	0.597
BOAMP:22-123056	de loire atlanti	IT Services & Consulting	0.376	0.592

Top buyers. Table 38 ranks buyers by expected 12-month renewal volume.

Table 38: Top 10 buyers by expected identifiable renewals in 12 months ($\sum_i \hat{p}_{i,12}$)

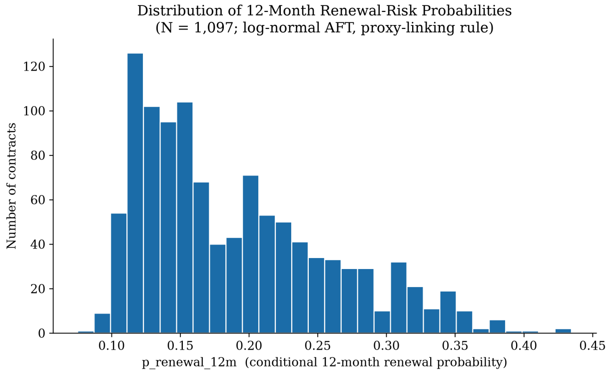
Buyer (normalised key)	n	$E[\text{ren.}]_{12}$	$E[\text{ren.}]_{24}$	mean p_{12}
des pays de la loire	85	18.58	32.04	0.219
nantes metropole	60	9.54	17.18	0.159
de nantes	39	6.48	11.52	0.166
loire atlantique	27	6.34	10.85	0.235
saint nazaire	34	6.05	10.74	0.178
angers loire metropole	29	5.79	10.07	0.200
le mans	22	5.20	8.81	0.236
gie sesam vitale	25	4.68	8.26	0.187
nantes	25	4.18	7.43	0.167
universite d angers	21	3.86	6.84	0.184

Segments. Table 39 gives the expected 12- and 24-month renewal volume by service segment.

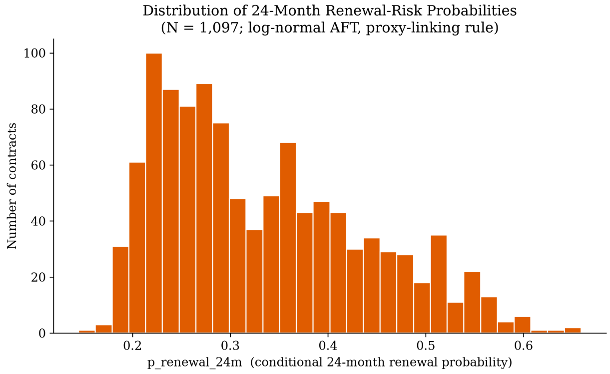
Table 39: Expected identifiable renewals by service segment (12 and 24 months)

Segment	n	$E[\text{ren.}]_{12}$	$E[\text{ren.}]_{24}$	mean p_{12}	mean p_{24}
IT Services & Consulting	312	61.61	107.23	0.198	0.344
Telecom & Networks	201	37.60	65.78	0.187	0.327
Software & Applications	209	37.49	66.11	0.179	0.316
Cybersecurity	121	23.77	41.44	0.196	0.343
Unknown	108	20.56	36.02	0.190	0.334
Digital Workplace & Collaboration	45	7.79	13.78	0.173	0.306
Data & AI	34	6.14	10.83	0.181	0.319
IT Maintenance & Support	22	4.58	7.92	0.208	0.360
Cloud & Infrastructure	19	3.94	6.82	0.208	0.359
IT Hardware & Equipment	21	3.78	6.67	0.180	0.318
GIS & Mapping	5	1.08	1.87	0.216	0.373

Distributions and rankings. Figure 21 shows the distribution of the 12- and 24-month renewal-risk probabilities; Figure 22 shows the top buyers and the segment-level expected renewal volume.

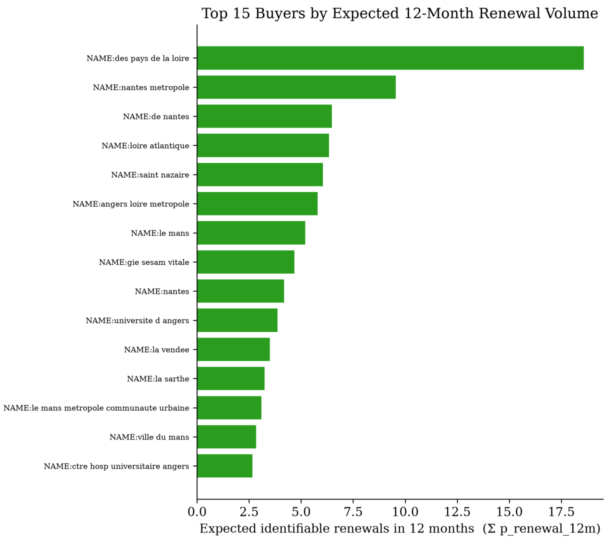


(a) 12-month probability $\hat{p}_{i,12}$

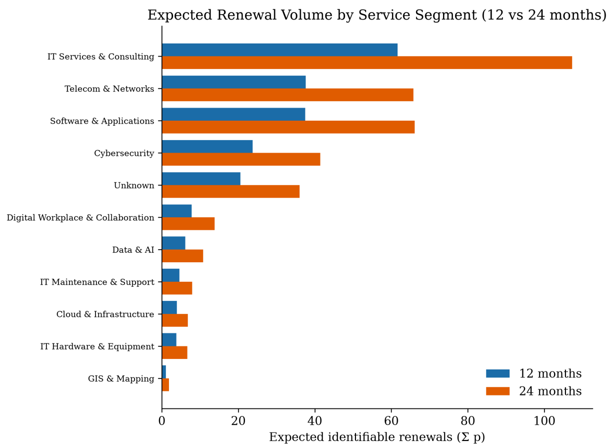


(b) 24-month probability $\hat{p}_{i,24}$

Figure 21: Distribution of contract-level renewal-risk probabilities under the log-normal AFT model (proxy-linking rule, $N = 1{,}097$).



(a) Top 15 buyers by $\sum_i \hat{p}_{i,12}$



(b) Expected renewals by segment (12 vs 24 mo)

Figure 22: Operational renewal-risk rankings: buyers with the largest near-term expected renewal volume (left) and expected renewal volume by service segment (right).

Limitation. These probabilities estimate the risk of an identifiable BOAMP renewal under the current proxy-linking rule. They should be interpreted as prioritization indicators, not as legally verified renewal probabilities. Manual validation of linked renewal pairs remains necessary to estimate the precision of the proxy event variable and to calibrate the operational risk scores.

14 Remaining Next Steps

Phase 2, including the operational renewal-risk prediction layer (Section 13.10), is complete. The remaining planned steps are:

1. **Manual validation:** Annotate a stratified sample of 50–100 linked pairs to estimate the precision of the proxy `event` variable.
2. **Threshold calibration:** Re-calibrate the operational risk-tier thresholds and prediction scores once the manual-validation sample is available.
3. **Trend / change-point analysis:** Analyse renewal-rate trends and structural breaks by technological segment.
4. **Optional DECP enrichment:** Revisit DECP integration only after the BOAMP-only model is stable and a defensible cross-source linkage strategy is defined. Exploratory artefacts already exist (`decp_renewal_links.csv`, `buyer_bridge.csv`, `unified_survival.csv`) but remain outside the validated BOAMP-only handoff.